

Theta Asymptotics for High-Power Ballot Sums

OEIS A357825, A357871, and the growing-power array A357824

Exact oscillation envelopes, all-orders expansions, exponential multiset corrections, and inverse asymptotics

Abstract

Let $B(n, h)$ count nonnegative unit-step paths of length n ending at height h , and let $a_n = \sum_h B(n, h)^n$. The OEIS records the leading root growth of this sequence and the failure of a natural normalized ratio to converge. We derive its missing multiplicative structure: the ratio is a Jacobi theta function evaluated at a drifting parity-sensitive square-root phase, times $e^{-5/6}$. A uniform all-orders expansion expresses every subsequent coefficient as a Gaussian-weighted polynomial sum, equivalently a finite linear combination of theta derivatives. We prove the exact lower and upper limiting constants, identify the entire cluster interval and limiting phase law, and obtain oscillatory adjacent-ratio and Turán asymptotics. In particular, the sequence is eventually strictly log-convex.

The multiset counterpart $b_n = \sum_h \binom{B(n, h) + n - 1}{n}$ has the same algebraic asymptotic expansion after multiplication by $n!$, but differs by an explicitly determined first exponential sector of relative order $n^3 2^{-n}$. An exact rising-factorial identity supplies every fixed exponential sector. We also treat the array $\sum_h B(n, h)^k$: its transition from a continuous Gaussian approximation to a finite-width lattice theta sum occurs at $k \asymp n$. An exact two-endpoint estimate covers all larger powers and yields a second, near-square logistic transition at $k \asymp n^{3/2}$. Finally, we derive parity-sensitive inverse expansions and explain their precise relation to integer thresholds. The proofs are accompanied by independent exact integer checks, symbolic coefficient generation, and high-precision numerical experiments.

Status and scope. This is a self-contained mathematical research report, not a Lean formalization or an independently refereed paper. The asymptotic formulas below are proved here; computational checks support but do not replace those proofs. No claim of established publication priority is made. The prime-power supercongruence recorded for A357825 is *not* resolved in this report.

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Part I

Theta asymptotics for high-power ballot sums

1 The question, the contribution, and its provenance

1.1 What the OEIS already records

For a nonnegative integer n and an admissible height $h \in \{n, n-2, \dots, n-2\lfloor n/2 \rfloor\}$, put

$$B(n, h) = \binom{n}{(n-h)/2} - \binom{n}{(n-h)/2 - 1}, \quad (1.1)$$

where a binomial coefficient with lower index -1 is zero. These are the ballot numbers in [A008315](#). Reflection at the first visit to height -1 gives the interpretation as nonnegative paths with steps $+1$ and -1 . Define

$$S_{n,k} = \sum_h B(n, h)^k, \quad a_n = S_{n,n}, \quad b_n = \sum_h \binom{B(n, h) + n - 1}{n}. \quad (1.2)$$

The array $S_{n,k}$ is [A357824](#), its diagonal a_n is [A357825](#), and b_n is [A357871](#) [4, 3, 1, 2]. The powers count ordered tuples sharing a common endpoint; the binomial coefficients count multisets sharing a common endpoint. The first values are

$$(a_n)_{n=0}^6 = (1, 1, 2, 9, 98, 4150, 562692), \quad (b_n)_{n=0}^6 = (1, 1, 2, 5, 21, 183, 3424).$$

For $n \geq 1$, set

$$K_n = \frac{2^{n+3/2}}{\sqrt{\pi e n}}, \quad \mathcal{A}_n = K_n^n = \frac{2^{n^2+3n/2}}{\pi^{n/2} e^{n/2} n^n}, \quad (1.3)$$

$$\mathcal{B}_n = \frac{e^{n/2} 2^{n^2+3n/2}}{\pi^{n/2} n^{2n+1/2}}. \quad (1.4)$$

Kotesovec's entries from November 2022 state $a_n^{1/n} \sim K_n$ and that $a_n/\mathcal{A}_n$ has no limit. The multiset entry gives the analogous root asymptotic and nonconvergence for $b_n/\mathcal{B}_n$ [1, 2]. Those observations are the starting point, not claims of this report's discovery. The natural next questions are: what replaces the missing constant, how large are the oscillations, and is there a uniform expansion beyond their leading shape?

1.2 What is established here

The main result is

$$\frac{a_n}{\mathcal{A}_n} = e^{-5/6} \left\{ \Theta(\alpha_n) + \frac{F_1(\alpha_n)}{\sqrt{n+1}} + \frac{F_2(\alpha_n)}{n+1} + \dots \right\}, \quad (1.5)$$

with a proved remainder after every fixed number of terms. Here

$$\alpha_n = \frac{\sqrt{n+1} - ((n+1) \bmod 2)}{2}, \quad \Theta(\alpha) = \sum_{j \in \mathbb{Z}} e^{-4(j-\alpha)^2}.$$

The coefficient functions are periodic and explicitly computable. The oscillations are therefore organized by a simple phase, not erratic noise. The main extensions are an exact limiting

interval and distribution, eventual log-convexity, the $k \asymp n$ transition for the entire array, all fixed exponential sectors connecting the tuple and multiset counts, and an inverse expansion retaining the same phase.

Question or assertion	Status in this report
Root growth and absence of a limit	Already recorded by OEIS; proved and sharpened here.
Theta factor and every algebraic order	Explicit formulas with a uniform localization proof.
Multiset exponential corrections	Exact finite identity and every fixed-sector remainder.
Growing powers and inverse formulas	Proved in the parameter ranges stated below.
Global log-convexity at every index	Not claimed; only eventual strict log-convexity is proved.
Bala's prime-power supercongruence	Left open; no arithmetic conclusion is inferred from asymptotics.
Lean verification and publication priority	Neither is claimed.

1.3 Relation to the literature and ProveIt

Catalan-triangle moments and fixed powers are part of an established literature; Miana and Romero study such moments, closed expressions in several cases, and divisibility questions [7]. The present asymptotic regime instead lets the exponent grow with the row index. Discrete Laplace methods are also established; a recent systematic formulation is given by Helton, Hughes, and Schlosser [8]. The point here is not to rename that method as new, but to retain the lattice when its mesh does *not* become small relative to the peak width. Standard Stirling and theta identities are used with attribution [9, 10, 11, 12].

Repository inspection used ProveIt commit

50f93367b2859b2727c34c68cf87684c101768cc

and its combinatorics map [13]. No existing repository theorem is required for the proofs below. This article is a proposed additional research case study; it does not promote a repository computation into a formal theorem. A repository code search for A357825 and targeted public searches for the sequence identifiers and theta/high-power ballot asymptotics did not identify a matching all-orders formula in the material reviewed. That is a limited search result, not a proof of novelty. The source-status audit and bibliography are dated 1 October 2026.

1.4 Place in the ProveIt collection

[Write note, batch 73O1, 1 October 2026.] This report is filed in ProveIt's research-report collection, in the directory

SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/
oeis-sequence-asymptotics/a357825-theta-ballot-power-sums.

Its README there is the guide to the files.

Provenance. The report is built from one manuscript: manuscript 59 of batch 73 (cluster O1) of ProveIt's incoming-reports intake, the archive OEIS_Theta_Ballot_Sums.zip, which

arrived in commit aa43cc555 and was placed by commit 9df4ba51a. Its pin is the commit 50f93367b quoted above (the delivered source audit calls it the “inspected tree commit”; it is a commit). The text is printed as delivered. The write changed only the label names, which now carry the prefix `tbs:`, and added the notes marked *[Write note]*. Nothing was merged into it, and no other manuscript of the batch treats these sequences.

Status. The manuscript was prepared with ChatGPT. It is unrefereed, and nothing in it is formalized in Lean or any other proof assistant. The intake reran the shipped verifier on a copy (`code/verify.py -all`, exit code 0; the eight data files it writes equal the delivered ones, up to line endings). It also checked independently the first terms of a_n and b_n , the two envelope constants of Theorem 4.2 to 20 digits, and the expansion through F_2 against exact sums for $n = 100, 200, 400, 800$. It did not re-derive every proof. The OEIS values and attributions quoted in the article were not re-fetched by the intake; they are as the manuscript inspected them on 1 October 2026.

Shipped layout. The delivered `requirements.txt` is shipped as `data/requirements.txt`, and `build.sh` as `code/build.sh`; the paths printed in Section 9 are the delivered ones. The figure PDFs were regenerated by the intake with embedded TrueType instead of Type 3 fonts, from the same data; see the README. The delivered README, PDF and checksum list are not shipped.

Neighbouring material. The same mechanism, a continuous saddle whose lattice is kept when its mesh stays comparable to the peak width, producing a theta multiplier, appears in two packages of ProveIt’s series-and-transseries collection, for different objects: the q -multinomial package `Theta_Resolved_Optimal_Truncation_q_Multinomial` and the chapter “Galois numbers and root-lattice theta sectors” of `Combinatorial_Transseries_Inverses`. They are cited for comparison only; no statement here depends on them, and none of theirs on this report. The report `hankel-determinants/catalan-and-ballot/ballot-polynomial-hankel-determinants` studies Hankel determinants of ballot moments, not growing powers. Bala’s supercongruence (question Q9 of Section 10) remains open.

[Write note, batch 98D, 5 October 2026.] Since batch 98 the report has two Parts. Part I is manuscript 59 of batch 73, as described above. Part II (Section 12) prints a later manuscript, the late batch-98 arrival `ProveIt_OEIS_Lattice_Peak_Transfer_2026-10-05.zip` (arrival 1ec443bc4, placement 3b9458b31), which answers question Q6 in a sufficient-hypothesis form: a general lattice-peak transfer theorem, with an all-orders, a multidimensional and an inverse version, of which Theorem 6.2 and the Mahonian report’s critical window are instances. Its labels carry the prefix `tbs:lpt:`, and its provenance is in Section 12.1. Nothing of Part I was changed except the Part heading and the notes dated 5 October 2026.

2 The shifted ballot identity and the main theorem

2.1 Why the shift by one matters

Put

$$m = n + 1, \quad r = h + 1, \quad \varepsilon_n = m \bmod 2 \in \{0, 1\}. \quad (2.1)$$

Then r ranges over the positive integers of parity m with $r \leq m$. A binomial simplification gives the exact identity

$$B(n, h) = \frac{r}{m} \binom{m}{(m-r)/2}. \quad (2.2)$$

Define the continuous positive interpolation, only as an analytic tool, by

$$b_m(r) = \frac{r}{m} \frac{\Gamma(m+1)}{\Gamma((m-r)/2+1)\Gamma((m+r)/2+1)}, \quad 0 < r \leq m, \quad (2.3)$$

and introduce

$$V_m = \frac{2^m}{m} \sqrt{\frac{2}{\pi e}} = K_n \frac{n}{m}. \quad (2.4)$$

The maximum of the leading continuous shape is at $r = \sqrt{m}$. Writing

$$\alpha_n = \frac{\sqrt{m} - \varepsilon_n}{2}, \quad u = r - \sqrt{m} = 2(j - \alpha_n), \quad (2.5)$$

turns the summation into a translated lattice of fixed spacing 2. The phase can be reduced modulo 1 whenever a periodic function is applied to it. Keeping $m = n + 1$, rather than replacing it prematurely by n , is necessary for the correct constant and correction terms.

Definition 2.1 (Theta sums and Gaussian moments). For $\tau > 0$ and $p \geq 0$, let

$$\Theta_\tau(\alpha) = \sum_{j \in \mathbb{Z}} e^{-4\tau(j-\alpha)^2}, \quad M_{p,\tau}(\alpha) = \sum_{j \in \mathbb{Z}} [2(j-\alpha)]^p e^{-4\tau(j-\alpha)^2}. \quad (2.6)$$

Write $\Theta = \Theta_1$ and $M_p = M_{p,1}$. These sums and all of their phase derivatives converge absolutely and locally uniformly in $\tau > 0$. They are smooth, 1-periodic functions of α .

Theorem 2.2 (All-orders diagonal expansion). *There are explicitly computable polynomials $R_j \in \mathbb{Q}[u]$, with $R_0 = 1$, $\deg R_j \leq 3j$, and $R_j(-u) = (-1)^j R_j(u)$, such that for every fixed $J \geq 0$,*

$$\frac{a_n}{\mathcal{A}_n} = e^{-5/6} \left\{ \sum_{j=0}^J m^{-j/2} F_j(\alpha_n) + \mathcal{O}_J(m^{-(J+1)/2}) \right\}, \quad m = n + 1, \quad (2.7)$$

where

$$F_j(\alpha) = \sum_{\ell \in \mathbb{Z}} e^{-u_\ell^2} R_j(u_\ell), \quad u_\ell = 2(\ell - \alpha). \quad (2.8)$$

The implied constant is independent of the phase. In particular $F_0 = \Theta$, and the same order holds as a relative error after division by the leading term.

The first two correction polynomials are

$$R_1(u) = \frac{u^3}{3} + \frac{2u}{3}, \quad (2.9)$$

$$R_2(u) = \frac{u^6}{18} - \frac{u^4}{36} + \frac{11u^2}{9} + \frac{13}{60}. \quad (2.10)$$

Consequently

$$F_1 = \frac{1}{3}M_3 + \frac{2}{3}M_1 = \frac{1}{192}\Theta''' + \frac{7}{24}\Theta', \quad (2.11)$$

$$F_2 = \frac{1}{18}M_6 - \frac{1}{36}M_4 + \frac{11}{9}M_2 + \frac{13}{60}\Theta. \quad (2.12)$$

All derivatives here are with respect to α . At integer and half-integer phases F_1 vanishes by symmetry. Away from those phases an $m^{-1/2}$ term is generally present; restricting an ansatz to integral powers of $1/n$ loses genuine information.

Remark 2.3 (Meaning of “all orders”). Equation (2.7) is a Poincaré asymptotic expansion: J is fixed while $n \rightarrow \infty$. No convergence of the infinite series, optimal truncation theorem, or resurgent description is asserted. Because its coefficients are periodic functions of a drifting phase, it is an *oscillatory* expansion, not an ordinary real logarithmic-exponential transseries with constant coefficients. The additional exponential sectors in Section 7 give a separate, precise beyond-all-orders statement.

3 Uniform localization and the proof of the expansion

3.1 A Stirling expansion with a moving argument

Define

$$f(x) = \log x - \frac{x^2 - 1}{2}, \quad x > 0. \quad (3.1)$$

It has a unique maximum 0 at $x = 1$, with $f''(1) = -2$. The logarithmic Stirling expansion for Γ gives the following uniform form.

Lemma 3.1 (Uniform local logarithm). *For every compact interval $I \subset (0, \infty)$ and every integer $L \geq 0$,*

$$\log \frac{b_m(\sqrt{m}x)}{V_m} = f(x) + \sum_{\ell=1}^L A_\ell(x) m^{-\ell} + O_{I,L}(m^{-L-1}), \quad x \in I, \quad (3.2)$$

for all sufficiently large m . The estimate may be differentiated any fixed number of times in x , after increasing the finite Stirling truncation used to establish it. Here

$$\begin{aligned} A_\ell(x) = & -\frac{x^{2\ell+2}}{(2\ell+2)(2\ell+1)} + \frac{x^{2\ell}}{2\ell} \\ & + \sum_{j=1}^{\lfloor (\ell+1)/2 \rfloor} \frac{B_{2j}}{2j(2j-1)} \left[\mathbf{1}_{\ell=2j-1} - 2^{2j} \binom{2\ell-2j}{2\ell-4j+2} x^{2\ell-4j+2} \right], \end{aligned} \quad (3.3)$$

where B_{2j} denotes a Bernoulli number, with $B_2 = 1/6$.

Proof. Set $z = x/\sqrt{m}$, and insert the logarithmic Stirling formula into (2.3). The exponential part of the binomial coefficient is controlled by

$$\frac{(1+z)\log(1+z) + (1-z)\log(1-z)}{2} = \sum_{q \geq 1} \frac{z^{2q}}{2q(2q-1)}.$$

Its $q = 1$ term gives $-x^2/2$, and its higher terms give the first term of (3.3). The square-root prefactor contributes $-\frac{1}{2} \log(1-z^2)$, producing the second term.

For a Bernoulli term with $p = 2j - 1$, subtraction of the two denominator corrections produces

$$\frac{B_{2j}}{2jp} m^{-p} [1 - 2^p \{(1-z)^{-p} + (1+z)^{-p}\}].$$

The even Taylor coefficient of z^{2k} in the braces is $2 \binom{p+2k-1}{2k}$. Setting $k = \ell - p$ yields the summation in (3.3). The leading prefactors, together with r/m , are exactly $V_m \exp f(x)$.

For x in a fixed compact interval, both denominator arguments are $m/2 + O(\sqrt{m})$, hence are bounded below by a positive multiple of m . Finite Stirling remainder bounds and finite Taylor

remainders are therefore uniform. The corresponding digamma and polygamma expansions supply the same conclusion after any fixed number of differentiations; see [9, 10]. This proves the finite asymptotic statement, without requiring convergence of the displayed infinite Taylor or Stirling series outside their use as coefficient generators. $\square$

For example,

$$\begin{aligned} A_1(x) &= -\frac{x^4}{12} + \frac{x^2}{2} - \frac{1}{4}, \\ A_2(x) &= -\frac{x^6}{30} + \frac{x^4}{4} - \frac{x^2}{3}, \\ A_3(x) &= -\frac{x^8}{56} + \frac{x^6}{6} - \frac{x^4}{3} + \frac{1}{24}. \end{aligned} \tag{3.4}$$

In particular $A_1(1) = 1/6$ and $A'_1(1) = 2/3$.

3.2 A global Gaussian bound

A local Stirling calculation alone does not justify exchanging an asymptotic expansion with a sum whose range grows. The necessary tail control follows from global concavity.

Lemma 3.2 (Uniform Gaussian domination). *There are constants $C, c > 0$ such that, for all sufficiently large n and all $0 < r \leq m = n + 1$,*

$$n \log \frac{b_m(r)}{K_n} \leq C - c(r - \sqrt{m})^2. \tag{3.5}$$

In particular the normalized summands of a_n have a phase-uniform summable Gaussian envelope.

Proof. Write $\ell_m(r) = \log b_m(r)$. Direct differentiation gives

$$\ell''_m(r) = -\frac{1}{r^2} - \frac{1}{4} \left[\psi_1\left(\frac{m-r}{2} + 1\right) + \psi_1\left(\frac{m+r}{2} + 1\right) \right], \tag{3.6}$$

where ψ_1 is the trigamma function. For $z > 0$ its positive series satisfies

$$\psi_1(z) = \sum_{q=0}^{\infty} (z+q)^{-2} \geq \int_0^{\infty} (z+t)^{-2} dt = \frac{1}{z}.$$

The two arguments in (3.6) sum to $m+2$, so their reciprocals sum to at least $4/(m+2)$. Thus $\ell''_m(r) \leq -1/(m+2)$ everywhere on the stated interval.

Lemma 3.1, including its differentiated form, gives

$$\ell'_m(\sqrt{m}) = \frac{2}{3m^{3/2}} + O(m^{-5/2}), \quad n \log \frac{b_m(\sqrt{m})}{K_n} = O(1).$$

Taylor's theorem with the global upper bound on the second derivative now gives, with $u = r - \sqrt{m}$,

$$n \log \frac{b_m(r)}{K_n} \leq C + O(m^{-1/2})u - \frac{n}{2(m+2)}u^2.$$

Completing the square absorbs the linear term and proves (3.5). $\square$

For $|u| > m^{1/12}$, the total contribution is

$$O\left(\exp(-c'm^{1/6})\right),$$

uniformly in the lattice translate, by the Gaussian envelope. This is smaller than every fixed algebraic power of m^{-1} . It also controls every fixed polynomial moment of the tails.

3.3 The coefficient engine

Let $t = m^{-1/2}$ and introduce a formal series in t whose coefficients are polynomials in u :

$$\begin{aligned}\mathcal{L}(t, u) &= -1 + (t^{-2} - 1) \left[f(1 + ut) + \sum_{\ell \geq 1} A_\ell (1 + ut) t^{2\ell} \right] + \sum_{\ell \geq 1} \frac{t^{2\ell}}{\ell(\ell + 1)} \\ &= -\frac{5}{6} - u^2 + \sum_{j \geq 1} D_j(u) t^j.\end{aligned}\quad (3.7)$$

Only finitely many terms are required to obtain any specified coefficient. The last sum appears because

$$(m - 1) \log(1 - 1/m) = -1 + \sum_{\ell \geq 1} \frac{m^{-\ell}}{\ell(\ell + 1)}.\quad (3.8)$$

Thus $\mathcal{L}$ generates precisely the expansion of $n \log(\mathfrak{b}_m(\sqrt{m} + u)/K_n)$. The first exponent polynomials are

$$\begin{aligned}D_1(u) &= \frac{u^3}{3} + \frac{2u}{3}, \\ D_2(u) &= -\frac{u^4}{4} + u^2 + \frac{13}{60}, \\ D_3(u) &= \frac{u^5}{5} - \frac{2u^3}{3} - \frac{8u}{15}, \\ D_4(u) &= -\frac{u^6}{6} + \frac{u^4}{6} + \frac{2u^2}{3} + \frac{59}{420}.\end{aligned}\quad (3.9)$$

Exponentiation gives the exact polynomial recurrence

$$R_0(u) = 1, \quad R_j(u) = \frac{1}{j} \sum_{i=1}^j i D_i(u) R_{j-i}(u) \quad (j \geq 1).\quad (3.10)$$

Indeed it is the coefficient recurrence for $\exp(\sum_{i \geq 1} D_i t^i) = \sum_{j \geq 0} R_j t^j$. Each D_i has parity i and degree at most $i + 2$. Any product contributing to weight j therefore has degree at most $3j$ and parity j , proving the structural assertions of Theorem 2.2.

The constant $-5/6$ is worth isolating. Raising the peak's Stirling correction to the n th power contributes $+1/6$ to the logarithm. Converting V_m^n to K_n^n contributes -1 . Their sum is $-5/6$; omitting either effect gives the wrong oscillation envelope.

3.4 A summable remainder and completion of the proof

Fix J . On $|u| \leq m^{1/12}$, apply Lemma 3.1 with a finite truncation larger than the order needed for t^{J+1} , and apply Taylor's theorem to the finitely many functions at $x = 1 + ut$. Since $|ut| \leq m^{-5/12}$, these Taylor expansions are uniform. The first nonconstant perturbation of the Gaussian exponent is bounded by $Ct(1 + |u|^3)$, which is $o(1)$ on this window. Finite Taylor's theorem for the exponential consequently gives some integer d_J and constants $C_J, c_J > 0$ such that

$$\begin{aligned}\left| \left(\frac{\mathfrak{b}_m(\sqrt{m} + u)}{K_n} \right)^n - e^{-5/6 - u^2} \sum_{j=0}^J m^{-j/2} R_j(u) \right| \\ \leq C_J m^{-(J+1)/2} (1 + |u|)^{d_J} e^{-c_J u^2}.\end{aligned}\quad (3.11)$$

To see why increasing the Stirling truncation suffices, its remainder before multiplication by n is $O(m^{-L-1})$ uniformly for x near 1; after multiplication it is $O(m^{-L})$. Choosing $L \geq \lceil (J+1)/2 \rceil$ and retaining the required Taylor terms makes it no larger than the requested order, with any extra polynomial factors absorbed into $(1+|u|)^{d_J}$. The finitely many exponentiation remainders have the same form.

For every $p \geq 0$ and $c > 0$,

$$\sup_{\alpha \in [0,1]} \sum_{j \in \mathbb{Z}} (1 + |2(j-\alpha)|)^p e^{-c[2(j-\alpha)]^2} < \infty.$$

Summing (3.11) over the translated lattice therefore preserves its algebraic order. Lemma 3.2 controls the omitted actual summands outside the window. The polynomial-Gaussian approximants have tails of the same negligible kind. Extending the finite admissible lattice to all of $\mathbb{Z}$ adds only such tails, since both endpoints are a growing distance from $\sqrt{m}$. This proves (2.7). Finally, the nearest lattice point has $|u| \leq 1$, so $\Theta(\alpha) \geq e^{-1}$ uniformly; absolute and relative versions of the theorem are equivalent. $\square$

4 Theta geometry, exact envelopes, and phase distribution

4.1 Theta identities and extrema

Poisson summation of a Gaussian gives

$$\Theta_\tau(\alpha) = \frac{\sqrt{\pi}}{2\sqrt{\tau}} \sum_{q \in \mathbb{Z}} \exp\left(-\frac{\pi^2 q^2}{4\tau}\right) e^{2\pi i q \alpha}. \quad (4.1)$$

This also follows from the standard theta transformation [12]. Termwise differentiation of (2.6) gives

$$M'_{p,\tau} = -2pM_{p-1,\tau} + 4\tau M_{p+1,\tau}. \quad (4.2)$$

For $\tau = 1$, iteration expresses every F_j as a finite rational linear combination of derivatives of Θ . This is an alternative all-orders representation to the rapidly convergent moment sums.

Lemma 4.1 (Strict extrema). *For each $\tau > 0$, Θ_τ is strictly decreasing on $(0, 1/2)$. Its global maximum is at integer phases and its global minimum is at half-integer phases.*

Proof. Let $q = \exp(-\pi^2/(4\tau)) \in (0, 1)$. Apart from the positive constant in (4.1), the Jacobi product is

$$\prod_{j \geq 1} (1 - q^{2j})(1 + 2q^{2j-1} \cos(2\pi\alpha) + q^{4j-2}).$$

Every phase-dependent factor is positive. Its logarithmic derivative is strictly negative when $0 < \alpha < 1/2$. The derivative series converges locally uniformly there, so the logarithmic derivative of the product is negative. Periodicity and reflection symmetry finish the proof. The product identity is standard [11]. $\square$

Theorem 4.2 (Exact limiting interval). *The cluster set of $a_n/\mathcal{A}_n$ is precisely $[C_-, C_+]$, where*

$$C_- = e^{-5/6} \sum_{j \in \mathbb{Z}} e^{-(2j-1)^2} = 0.31986675953099572612\dots, \quad (4.3)$$

$$C_+ = e^{-5/6} \sum_{j \in \mathbb{Z}} e^{-4j^2} = 0.45051819401966197435\dots \quad (4.4)$$

In particular these are respectively the liminf and limsup, and no constant C satisfies $a_n \sim C\mathcal{A}_n$.

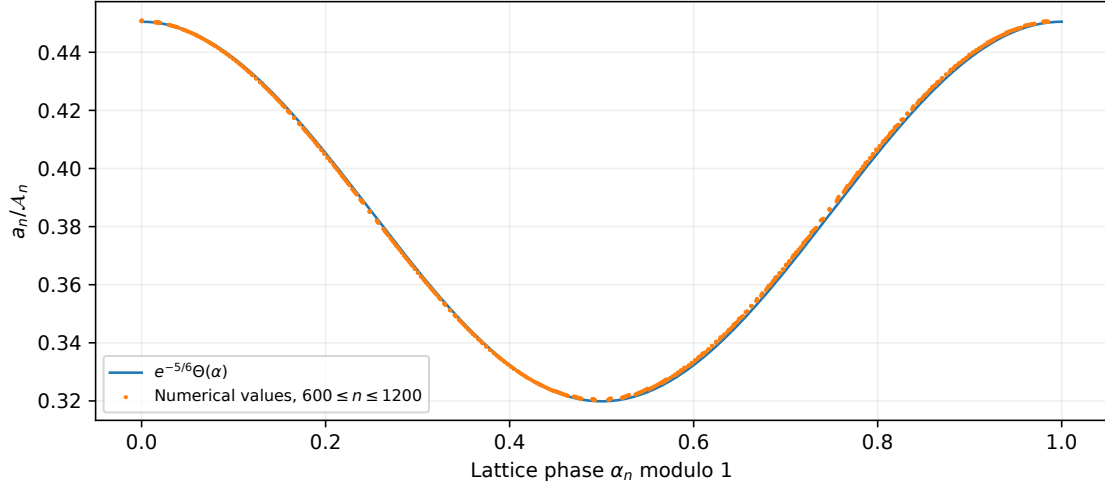


Figure 1: The apparent oscillation collapses onto a single theta curve when plotted against the phase. Dots are high-precision localized log-Gamma evaluations for $600 \leq n \leq 1200$, not interval-certified values. The discrepancy is explained by the $m^{-1/2}$ and higher terms.

Proof. Theorem 2.2 reduces the claim to the range of Θ along the phases. Lemma 4.1 supplies universal limiting bounds. For $n = s^2 - 1$, the phase is an integer exactly, so this subsequence converges to C_+ . For $n = s^2$, the phase tends modulo one to $1/2$, so that subsequence converges to C_- . Proposition 4.3 below proves that every phase occurs as a subsequential limit. Continuity of Θ then fills the entire interval and excludes other cluster values. $\square$

4.2 An elementary equidistribution proof

Proposition 4.3 (Effective phase distribution). *For every interval $I \subset [0, 1)$,*

$$\#\{1 \leq n \leq N : \{\alpha_n\} \in I\} = N|I| + O(\sqrt{N}). \quad (4.5)$$

The implied constant can be chosen independently of I . The same assertion holds on either parity class, with its own number of terms in place of N .

Proof. Fix $\varepsilon \in \{0, 1\}$ and restrict $m = n + 1$ to that parity. For $I = [a, b)$, the condition is equivalent to membership in the union of intervals

$$(2j + \varepsilon + 2a)^2 \leq m < (2j + \varepsilon + 2b)^2,$$

apart from finitely many harmless initial endpoints. The number of integers of a specified parity in an interval of length L is $L/2 + O(1)$. The displayed interval has length

$$8j(b - a) + 4\varepsilon(b - a) + 4(b^2 - a^2).$$

There are $O(\sqrt{N})$ complete intervals up to $N + 1$. Summing their main terms gives $(b - a)$ times the number of admissible m up to $N + 1$, with error $O(\sqrt{N})$: the terms linear in j produce the quadratic main term, and all endpoint and linear-size corrections are of order $\sqrt{N}$. A final partial interval also has length $O(\sqrt{N})$ and obeys the same bound. All constants are uniform for $0 \leq a \leq b \leq 1$. Summing the two parity classes proves (4.5). $\square$

Corollary 4.4 (Limiting law and mean). *If an index is chosen uniformly from $1, \dots, N$, then $a_n/\mathcal{A}_n$ converges in distribution to $e^{-5/6}\Theta(U)$, where U is uniform on $[0, 1]$. Moreover*

$$\frac{1}{N} \sum_{n=1}^N \frac{a_n}{\mathcal{A}_n} = e^{-5/6} \frac{\sqrt{\pi}}{2} + O(N^{-1/2}) = 0.38515263413250459606 \dots + O(N^{-1/2}). \quad (4.6)$$

Proof. Continuity and equidistribution prove convergence in distribution. Integrating the discrepancy bound against the smooth function Θ gives an $O(N^{-1/2})$ error in its average. The average of the uniform $O(n^{-1/2})$ remainder in Theorem 2.2 has the same order. Finally

$$\int_0^1 \Theta(\alpha) d\alpha = \frac{1}{2} \int_{-\infty}^{\infty} e^{-u^2} du = \frac{\sqrt{\pi}}{2}.$$

□

The root asymptotic in the OEIS follows immediately: $a_n/\mathcal{A}_n$ is bounded above and bounded away from zero, so $a_n^{1/n}/K_n = (a_n/\mathcal{A}_n)^{1/n} \rightarrow 1$. The multiplicative oscillation survives in the ratio but disappears under the n th root.

5 Adjacent quotients, log-convexity, and recurrence obstructions

Put

$$\lambda = \log 2, \quad c = \frac{3}{2} \log 2 - \frac{1 + \log \pi}{2}, \quad g(x) = \lambda x^2 - x \log x + cx. \quad (5.1)$$

Then $g(n) = \log \mathcal{A}_n$. Consecutive phases satisfy

$$\alpha_{n\pm 1} = \alpha_n + \frac{1}{2} + O(n^{-1/2}) \pmod{1}. \quad (5.2)$$

Thus averaging away the phase before forming an adjacent quotient would lose an order-one factor.

Theorem 5.1 (Adjacent and Turán quotients). *As $n \rightarrow \infty$,*

$$\frac{a_{n+1}}{a_n} = \frac{2^{2n+5/2}}{\sqrt{\pi} e^{3/2} n} \frac{\Theta(\alpha_n + 1/2)}{\Theta(\alpha_n)} (1 + O(n^{-1/2})), \quad (5.3)$$

$$\frac{a_{n-1}a_{n+1}}{a_n^2} = 4 \left(\frac{\Theta(\alpha_n + 1/2)}{\Theta(\alpha_n)} \right)^2 + O(n^{-1/2}). \quad (5.4)$$

The second quotient has lower and upper limits

$$T_- = 4 \left(\frac{\Theta(1/2)}{\Theta(0)} \right)^2 = 2.01638539749423251125 \dots, \quad (5.5)$$

$$T_+ = 4 \left(\frac{\Theta(0)}{\Theta(1/2)} \right)^2 = 7.93499100910135656311 \dots \quad (5.6)$$

In particular a_n is strictly log-convex for all sufficiently large n .

Proof. Taylor expansion of g gives

$$e^{g(n+1)-g(n)} = \frac{2^{2n+5/2}}{\sqrt{\pi} e^{3/2} n} (1 + O(n^{-1})), \quad e^{g(n-1)+g(n+1)-2g(n)} = 4(1 + O(n^{-1})).$$

Use Theorem 2.2, (5.2), smoothness, and positivity of Θ . The ratio of two theta values is bounded by $\Theta(1/2)/\Theta(0)$ and its reciprocal. Both bounds are attained at phases 0 and $1/2$, which occur as subsequential limits.

The strict inequality $T_- > 1$ need not rest on floating-point decimals. For instance $\Theta(1/2) \geq 2/e > 2/3$, while

$$\Theta(0) = 1 + 2 \sum_{j \geq 1} e^{-4j^2} \leq 1 + \frac{2e^{-4}}{1 - e^{-12}} < \frac{11}{10}.$$

The last elementary bound follows already from $e > 8/3$. Consequently $T_- > 4(20/33)^2 > 1$, a positive gap from the log-convexity threshold. Equation (5.4) proves the final assertion. $\square$

A rapidly growing sequence need not automatically be log-convex: oscillatory multipliers can interfere with a second difference. Here the theta quotient is explicitly controlled, so the conclusion has a genuine proof rather than following from growth alone. The theorem does not specify the least index from which log-convexity holds.

Proposition 5.2 (No polynomial-coefficient linear recurrence). *Neither (a_n) nor (b_n) is P -recursive over $\mathbb{C}$. Equivalently, neither ordinary generating series is D -finite as a formal power series. Both ordinary generating series have radius of convergence zero.*

Proof. A sequence u_n satisfying a nonzero polynomial-coefficient recurrence $\sum_{j=0}^r p_j(n)u_{n+j} = 0$ can, for all large n , be solved for the term with largest nonzero index. Ratios of its polynomial coefficients are bounded in magnitude by Cn^d for some integer $d \geq 0$. If $M_N = \max(1, |u_0|, \dots, |u_N|)$, it follows that $M_{N+1} \leq C'N^d M_N$ after adjusting constants and ignoring a finite initial segment. Hence

$$|u_n| \leq C_0 C_1^n (n!)^d, \quad \log^+ |u_n| = O(n \log n).$$

A recurrence of order zero would force eventual vanishing and is also impossible here. Theorem 2.2 gives $\log a_n = \lambda n^2 + O(n \log n)$. Theorem 7.1 below gives the same quadratic leading logarithm for b_n . Both contradict the recurrence bound. The equivalence with formal D -finiteness follows by coefficient extraction from a linear differential equation and the converse conversion of a recurrence to such an equation. Finally $a_n^{1/n}$ and $b_n^{1/n}$ tend to infinity, so the radius assertion follows from the root test. $\square$

6 The growing-power array and two localization transitions

The scale of the exponent determines whether the lattice can be replaced by an integral. This section treats $S_{n,k}$ for real $k > 0$ analytically; integer k gives the OEIS array. The parameter $m = n + 1$ is always retained.

6.1 Fixed powers and the broad Gaussian regime

Proposition 6.1 (Fixed and subcritical powers). *For each fixed $k > 0$,*

$$S_{n,k} \sim V_m^k \frac{\sqrt{m}}{4} e^{k/2} \left(\frac{2}{k}\right)^{(k+1)/2} \Gamma\left(\frac{k+1}{2}\right). \quad (6.1)$$

If $k = k(n) \rightarrow \infty$ and $k = o(m)$, then

$$S_{n,k} \sim V_m^k \frac{\sqrt{\pi m/k}}{2}. \quad (6.2)$$

Proof. For fixed k , use $x = r/\sqrt{m}$. Its lattice mesh is $2/\sqrt{m}$. Lemma 3.1 gives pointwise convergence of $(b_m(\sqrt{m}x)/V_m)^k$ to $e^{kf(x)} = e^{k/2}x^k e^{-kx^2/2}$. The concavity proof gives a uniform integrable Gaussian bound in $x - 1$; compact-interval convergence and this tail bound justify the Riemann sum. Thus

$$S_{n,k}/V_m^k \sim \frac{\sqrt{m}}{2} \int_0^\infty e^{k/2}x^k e^{-kx^2/2} dx,$$

whose integral is (6.1).

For growing $k = o(m)$, use the finer local variable $v = (r - \sqrt{m})\sqrt{k/m}$. The mesh $2\sqrt{k/m}$ tends to zero, while

$$kf(1 + v/\sqrt{k}) \longrightarrow -v^2, \quad kA_1(1 + v/\sqrt{k})/m \longrightarrow 0$$

uniformly for bounded v . The same concavity argument gives a uniform Gaussian tail in v . The Riemann-sum limit is therefore $\frac{1}{2}\sqrt{m/k} \int_{\mathbb{R}} e^{-v^2} dv$, proving (6.2). $\square$

For orientation, the first two integral powers also have exact familiar identities:

$$S_{n,1} = \binom{n}{\lfloor n/2 \rfloor}, \quad S_{n,2} = \frac{1}{n+1} \binom{2n}{n}.$$

The first telescopes in (1.1). For the second, join a pair of nonnegative n -step paths with common height by following the first and then the reverse of the second. This is a bijection with Dyck paths of length $2n$. These columns are already identified by the OEIS [3]; they are consistency checks, not new enumeration claims.

6.2 The critical theta regime

Theorem 6.2 (Critical growing powers). *Let $\tau = k/m$ stay in a compact subinterval of $(0, \infty)$. Uniformly in τ and in the phase,*

$$S_{n,k} = V_m^k e^{\tau/6} [\Theta_\tau(\alpha_n) + O(m^{-1/2})]. \quad (6.3)$$

There is an expansion to every fixed order in $m^{-1/2}$, with coefficients that are polynomial-Gaussian sums and smooth in τ . Its first correction inside the brackets is

$$\frac{\tau}{\sqrt{m}} \sum_{j \in \mathbb{Z}} \left(\frac{u_j^3}{3} + \frac{2u_j}{3} \right) e^{-\tau u_j^2}, \quad u_j = 2(j - \alpha_n). \quad (6.4)$$

Proof. Replace the factor $n = m - 1$ in the local logarithmic analysis by $k = \tau m$. Lemma 3.1 gives

$$k \log \frac{b_m(\sqrt{m} + u)}{V_m} = \tau \left(\frac{1}{6} - u^2 \right) + \frac{\tau}{\sqrt{m}} \left(\frac{u^3}{3} + \frac{2u}{3} \right) + O(m^{-1}(1 + |u|)^4)$$

on bounded windows, uniformly for τ in the given compact interval. All higher coefficients follow by the same finite Taylor procedure. The global concavity bound, now multiplied by k , gives a summable Gaussian with constants uniform on that interval. The argument of (3.11) therefore applies without change. $\square$

The width in r is of order $\sqrt{m/k}$. It is large in (6.2), but of order one in (6.3). This explains the first localization transition, $k \asymp m$. [Write note, batch 98D, 5 October 2026: the leading term of Theorem 6.2 is an instance of Part II's general transfer theorem (Theorem 15.1 with Proposition 20.1), with $q = 8\tau$.]

For a probabilistic formulation, choose an endpoint with probability $\mathbb{P}_{n,k}(h) = B(n, h)^k / S_{n,k}$. After writing $u = h + 1 - \sqrt{m}$, the critical law is approximated in total variation, with error $O(m^{-1/2})$, by the phase-shifted discrete Gaussian

$$\mathbb{P}\{u = 2(j - \alpha_n)\} = \frac{e^{-4\tau(j-\alpha_n)^2}}{\Theta_\tau(\alpha_n)}. \quad (6.5)$$

Here the comparison extends the admissible finite lattice to all integers; the added tail is negligible. The weighted remainder bound in the proof also yields

$$\mathbb{E}_{n,k}[h + 1 - \sqrt{m}] = \frac{\Theta'_\tau(\alpha_n)}{4\tau\Theta_\tau(\alpha_n)} + O(m^{-1/2}). \quad (6.6)$$

The limit is not a single continuous Gaussian: subsequences with different phases have different discrete endpoint laws.

6.3 Exact modes and a uniform two-endpoint theorem

An exact ratio gives more than a truncated Stirling approximation in the supercritical regime. From (2.2),

$$\rho_m(r) := \frac{(r+2)(m-r)}{r(m+r+2)}, \quad 0 < r < m. \quad (6.7)$$

It equals $b_m(r+2)/b_m(r)$ when both arguments lie in $(0, m]$, and equals 1 exactly when $m = r(r+2)$.

Theorem 6.3 (Exact modes and all supercritical powers). *For $m \geq 3$, set $y = \sqrt{m+1}$, and let r_- be the largest positive integer of parity m satisfying $r_- \leq y$. Put $r_+ = r_- + 2$ and $W_\pm = b_m(r_\pm)$.*

The maximum ballot entry occurs at the admissible r nearest to y . There are exactly two tied maxima if and only if $m+1 = s^2$ for an integer $s \geq 2$; then their locations are $r = s-1, s+1$. For every real $k > 0$,

$$0 \leq \frac{S_{n,k}}{W_-^k + W_+^k} - 1 \leq \sum_{\ell \geq 1} \exp\left(-\frac{2k\ell^2}{m+1}\right). \quad (6.8)$$

Consequently, whenever $k/m \rightarrow \infty$, with no upper bound on k ,

$$S_{n,k} = (W_-^k + W_+^k) \left[1 + O\left(e^{-2k/(m+1)}\right) \right], \quad (6.9)$$

uniformly in the phase.

Proof. The sign of $\rho_m(r) - 1$ is the sign of $m - r(r+2)$, so adjacent entries increase up to the midpoint criterion $r+1 = y$ and decrease after it. This proves the nearest-point and exact-tie statements. The candidate pair lies within $0 < r \leq m$ for $m \geq 3$.

For the uniform remainder, differentiate the logarithm of the exact ratio:

$$\begin{aligned} \frac{d}{dr} \log \rho_m(r) &= -\frac{2}{r(r+2)} - \frac{1}{m-r} - \frac{1}{m+r+2} \\ &\leq -\frac{2}{m+1}, \quad 0 < r < m. \end{aligned}$$

The last inequality follows because $(m - r)(m + r + 2) = (m + 1)^2 - (r + 1)^2 \leq (m + 1)^2$. Also $\log \rho_m(y - 1) = 0$.

For the i th ratio to the left of r_- , the distance from its argument $r_- - 2i$ to $y - 1$ is at least $2i - 1$. Integrating the derivative bound and summing the ratios therefore gives

$$\frac{b_m(r_- - 2\ell)}{W_-} \leq \exp\left(-\frac{2\ell^2}{m + 1}\right)$$

whenever that lattice point is admissible. The ratios to the right of r_+ obey the same estimate relative to W_+ . Raising to k and summing both tails proves (6.8). Finally, with $t = k/(m + 1)$,

$$\sum_{\ell \geq 1} e^{-2t\ell^2} \leq \frac{e^{-2t}}{1 - e^{-6t}},$$

which proves the claimed supercritical order as $t \rightarrow \infty$. $\square$

This theorem keeps the two dominant ballot entries *exact*. It is therefore valid even when errors from raising a Stirling approximation to an extremely large power would be unacceptable. It also identifies a small but consequential distinction: the leading continuous saddle is $\sqrt{m}$, whereas the exact discrete mode is selected by $\sqrt{m + 1}$. A shift negligible at $k \asymp m$ need not be negligible at a much larger exponent.

The two main terms can equivalently be written

$$W_-^k + W_+^k = 2(W_- W_+)^{k/2} \cosh\left(\frac{k}{2} \log \frac{W_+}{W_-}\right). \quad (6.10)$$

Thus the precise crossover is governed by $k \log(W_+/W_-)$, not by an uncontrolled choice of one approximate maximum.

6.4 A second transition near square-index ties

Corollary 6.4 (Near-square logistic limit). *Fix an integer d and $\kappa > 0$. Along*

$$n = s^2 - 2 + 2d, \quad k/s^3 \rightarrow \kappa, \quad s \rightarrow \infty, \quad (6.11)$$

the endpoint law concentrates on $h = s - 2$ and $h = s$, and

$$\mathbb{P}_{n,k}\{h = s\} \rightarrow \frac{1}{1 + e^{-4\kappa d}}, \quad \mathbb{P}_{n,k}\{h = s - 2\} \rightarrow \frac{1}{1 + e^{4\kappa d}}. \quad (6.12)$$

For $k = \kappa s^3 + O(1)$ the errors are $O(s^{-2})$.

Proof. For fixed d and large s , the bracketing admissible points are $r_- = s - 1$ and $r_+ = s + 1$. The factor $2d$ in (6.11) preserves their parity; replacing it by an arbitrary odd shift would change the endpoint lattice. Substitution into (6.7) gives the exact identity

$$\frac{W_+}{W_-} = 1 + \frac{4d}{(s - 1)(s^2 + s + 2d)}, \quad \log \frac{W_+}{W_-} = \frac{4d}{s^3} + O_d(s^{-5}). \quad (6.13)$$

Since $k/m \sim \kappa s \rightarrow \infty$, Theorem 6.3 makes the other endpoints exponentially negligible. The upper endpoint probability is therefore $\rho^k/(1 + \rho^k)$ plus an exponentially small error. Equation (6.13) yields (6.12) and its error bound. $\square$

The first transition, at $k \asymp n$, changes a broad Gaussian sum into a finite-width lattice sum. The second, at $k \asymp n^{3/2}$ near square-index ties, resolves two neighboring entries whose logarithms differ by only order $n^{-3/2}$. At $d = 0$ the exact tie persists and the two limiting probabilities are $1/2$. This second transition is not an assumption about the phase: it follows from the exact ballot ratio.

7 Multisets and exponentially small sectors

7.1 An exact bridge between the two sequences

Let $E_q(n) = e_q(0, 1, \dots, n-1)$ be an elementary symmetric polynomial, so that

$$\prod_{j=0}^{n-1} (X + j) = \sum_{q=0}^n E_q(n) X^{n-q}. \quad (7.1)$$

In particular $E_0 = 1$, $E_1 = \binom{n}{2}$, and $E_n = 0$ for $n \geq 1$. Multiplying each multiset binomial by $n!$ and summing yields the exact identity

$$n!b_n = \sum_{q=0}^n E_q(n) S_{n,n-q}. \quad (7.2)$$

This identity separates the dominant tuple count from its successive corrections without approximating individual small ballot entries.

Theorem 7.1 (Every fixed exponential sector). *For each fixed $Q \geq 0$, as $n \rightarrow \infty$,*

$$\frac{n!b_n}{a_n} = \sum_{q=0}^Q E_q(n) \frac{S_{n,n-q}}{a_n} + O_Q\left(n^{2Q+2} V_m^{-Q-1}\right). \quad (7.3)$$

For each fixed q , the quotient $S_{n,n-q}/a_n$ has a phase-uniform expansion to every algebraic order, with leading term V_m^{-q} . Thus (7.3) is an expansion in successive exponentially small scales, each with its own algebraic expansion.

Proof. Nonnegativity and the multinomial expansion give

$$0 \leq E_q(n) \leq \frac{E_1(n)^q}{q!}.$$

For $0 \leq q \leq n/2$, Theorems 2.2 and 6.2, uniformly for the exponent ratio in a fixed interval around $[1/2, 1]$, imply

$$\frac{S_{n,n-q}}{a_n} \leq C V_m^{-q}.$$

For the remaining indices, let M be the largest ballot entry. The nearest-point estimate and the local expansion give $M/V_m = \exp(O(1/m))$. Since there are at most $n+1$ summands and $a_n \geq M^n$,

$$\frac{S_{n,n-q}}{a_n} \leq (n+1)M^{-q} \leq Cn V_m^{-q}.$$

Put $x = E_1/V_m$, which tends to zero exponentially. Summing the first bound over $q \geq Q+1$ through $n/2$ gives $O_Q(x^{Q+1})$. The remaining tail is at most $Cn \sum_{q>n/2} x^q/q!$, smaller than every fixed power of x for large n . The exact identity (7.2) proves the remainder.

For fixed q , apply the all-orders form of Theorem 6.2 with $\tau = (n - q)/m = 1 - (q + 1)/m$ and Taylor-expand its smooth coefficient functions in τ . Divide by the positive all-orders denominator from Theorem 2.2. Formal series division is justified at each finite order because the leading theta factor is bounded below. The leading factors cancel except for V_m^{-q} . $\square$

7.2 The first sector and its phase correction

Theorem 7.2 (Explicit leading multiset correction). *Let*

$$\Lambda(\alpha) = \frac{M_2(\alpha)}{\Theta(\alpha)} - \frac{1}{6}. \quad (7.4)$$

Then

$$\frac{n!b_n}{a_n} = 1 + \frac{\binom{n}{2}}{V_m} \left[1 + \frac{\Lambda(\alpha_n)}{m} + O(m^{-3/2}) \right] + O(n^4/V_m^2). \quad (7.5)$$

In particular,

$$\boxed{\frac{n!b_n}{a_n} = 1 + \frac{\sqrt{\pi e}}{4\sqrt{2}} n^3 2^{-n} (1 + O(n^{-1}))}. \quad (7.6)$$

Proof. Under the probability measure proportional to $B(n, h)^n$, the ratio $S_{n,n-1}/a_n$ is the expectation of $B(n, h)^{-1}$. The local logarithmic expansion gives

$$\log \frac{b_m(\sqrt{m} + u)}{V_m} = \frac{1/6 - u^2}{m} + O\left(m^{-3/2}(1 + |u|)^3\right)$$

in the summable expansion sense proved in Section 3. Consequently

$$\frac{S_{n,n-1}}{a_n} = V_m^{-1} \left[1 + \frac{M_2(\alpha_n)/\Theta(\alpha_n) - 1/6}{m} + O(m^{-3/2}) \right].$$

There is no $m^{-1/2}$ term in this quotient: the reciprocal ballot entry differs from V_m^{-1} first at order m^{-1} , and the $m^{-1/2}$ change in its weighting affects the expectation only at order $m^{-3/2}$. Polynomial-weighted Gaussian tail control justifies this multiplication and expectation.

Take $Q = 1$ in Theorem 7.1. Finally,

$$\frac{\binom{n}{2}}{V_m} = \frac{\sqrt{\pi e}}{4\sqrt{2}} (n^3 - n) 2^{-n}.$$

The boundedness of Λ and the smaller next sector yield (7.6). $\square$

This proves that $b_n \sim a_n/n!$ with an exponentially small relative error, not merely with an unspecified $o(1)$ error. It also prevents a misleading conclusion: although the two sequences share every algebraic asymptotic order after multiplication by $n!$, they are not identical beyond all orders.

7.3 The OEIS normalization for the multiset sequence

The ordinary reciprocal Stirling expansion gives

$$\frac{\mathcal{A}_n}{n!} = \frac{\mathcal{B}_n}{\sqrt{2\pi}} \left[1 - \frac{1}{12n} + O(n^{-2}) \right]. \quad (7.7)$$

Combining this with the diagonal expansion and the exponential estimate shows

$$\frac{b_n}{\mathcal{B}_n} = \frac{e^{-5/6}}{\sqrt{2\pi}} \left[\Theta(\alpha_n) + m^{-1/2} F_1(\alpha_n) + m^{-1} \left(F_2(\alpha_n) - \frac{\Theta(\alpha_n)}{12} \right) + O(m^{-3/2}) \right]. \quad (7.8)$$

An all-orders version follows by multiplication of the two finite asymptotic expansions. Its exact limiting interval is $[C_-/\sqrt{2\pi}, C_+/\sqrt{2\pi}]$, and its normalized limiting law is that in Corollary 4.4 scaled by $1/\sqrt{2\pi}$. This proves and refines the nonconvergence recorded for A357871. It also gives

$$b_n^{1/n} \sim \frac{e^{1/2} 2^{n+3/2}}{\sqrt{\pi} n^2},$$

agreeing with the existing root formula [2].

Finally, the Turán quotient for b_n differs from that of a_n by $n/(n+1)$ times an exponentially small relative perturbation. Its liminf and limsup are therefore also T_- and T_+ . In particular b_n is eventually strictly log-convex as well.

8 Inverting the oscillatory expansion

8.1 Why an inverse must specify a parity branch

An integer sequence does not come with a canonical continuous inverse. Moreover α_n contains a parity switch, so interpolating its two parity classes by a single differentiable phase is inappropriate. We instead define explicit asymptotic models and quantify how accurately their inverses recover the original integer data.

For $\sigma \in \{0, 1\}$ and real $x > 0$, set

$$\alpha_\sigma(x) = \frac{\sqrt{x+1} - \sigma}{2}. \quad (8.1)$$

The branch appropriate to an integer n has $\sigma = (n+1) \bmod 2$. For each fixed $J \geq 0$, define

$$A_{\sigma,J}(x) = \exp(g(x)) e^{-5/6} \sum_{j=0}^J (x+1)^{-j/2} F_j(\alpha_\sigma(x)). \quad (8.2)$$

This is a definition of a smooth approximating function, not a claim that the exact counting sequence has that interpolation.

Proposition 8.1 (Matched inverse accuracy). *For each fixed J and σ , $A_{\sigma,J}$ is positive and strictly increasing for all sufficiently large x . If $r_{\sigma,J}(L)$ is the large solution of $\log A_{\sigma,J}(x) = L$, then, for integers n of the matching parity,*

$$r_{\sigma,J}(\log a_n) - n = O_J(n^{-(J+3)/2}). \quad (8.3)$$

Proof. Positivity follows from the positive lower bound on Θ and the boundedness of all coefficient functions. Differentiating (8.2), and using $\alpha'_\sigma(x) = 1/(4\sqrt{x+1})$, gives

$$\frac{d}{dx} \log A_{\sigma,J}(x) = 2\lambda x - \log x - 1 + c + O(x^{-1/2}) = 2\lambda x + O(\log x) > 0.$$

Theorem 2.2 gives a logarithmic residual at $x = n$ of order $n^{-(J+1)/2}$. The mean value theorem divides that residual by a quantity comparable to n , proving (8.3). $\square$

Define the first logarithmic coefficient functions

$$d_0(\alpha) = -\frac{5}{6} + \log \Theta(\alpha), \quad d_1(\alpha) = \frac{F_1(\alpha)}{\Theta(\alpha)}. \quad (8.4)$$

Thus

$$\log a_n = g(n) + d_0(\alpha_n) + m^{-1/2}d_1(\alpha_n) + O(m^{-1}).$$

Higher logarithmic coefficients are obtained by ordinary finite series expansion of the logarithm; for example $d_2 = F_2/\Theta - F_1^2/(2\Theta^2)$.

8.2 An explicit inverse through the first oscillatory orders

Theorem 8.2 (Parity-sensitive inverse expansion). *Let $L \rightarrow \infty$, and write*

$$s = \sqrt{\frac{L}{\lambda}}, \quad d = \frac{\log s - c}{2\lambda}, \quad \beta = \alpha_\sigma(s + d). \quad (8.5)$$

For any fixed $J \geq 1$,

$$r_{\sigma,J}(L) = s + d + \frac{\lambda d^2 + d - d_0(\beta)}{2\lambda s} - \frac{d_1(\beta)}{2\lambda s^{3/2}} + O\left(\frac{(\log s)^2}{s^2}\right). \quad (8.6)$$

The implied constant is uniform in the two parity branches.

Proof. Since $\lambda s^2 = L$ and $2\lambda d = \log s - c$, Taylor expansion gives

$$g(s + d) - L = -\lambda d^2 - d + O(d^2/s), \quad g'(s + d) = 2\lambda s - 1 + O(d/s).$$

The logarithm of the model at $s + d$ adds $d_0(\beta) + s^{-1/2}d_1(\beta) + O((1 + d)/s)$. A Newton correction using the leading derivative $2\lambda s$ is exactly the two displayed correction terms in (8.6). Its remaining logarithmic residual is $O((1 + d)^2/s)$, including the derivative correction and the omitted model terms. Phase drift during this correction is bounded by the correction times $O(s^{-1/2})$ and is smaller than this residual. Dividing once more by a derivative comparable to s proves the stated error, because $d = O(\log s)$. $\square$

The phase in this formula is evaluated at $s + d$, not merely at s . That choice incorporates the logarithmic shift before applying the oscillatory correction. Dropping $d_0(\beta)$ discards an order- s^{-1} term; replacing it by its average cannot recover individual inverse values at that order.

For any requested finite accuracy, compute more F_j , form the logarithm of (8.2), and apply Newton iteration or formal coefficient reversion. At large x , the derivative is of order x and the second derivative is bounded. Hence a Newton step with an input error $e = o(x)$ has output error $O(e^2/x)$ relative to the chosen model root. Along with Proposition 8.1, this gives an effective all-orders inversion procedure. It does not assert convergence of an infinite formal inverse expansion.

8.3 Integer thresholds and the multiset inverse

For $y > 1$, let

$$N(y) = \min\{n \geq 1 : a_n \geq y\}.$$

The sequence is strictly increasing from $n = 1$ onward. One direct injection extends every path in an ordered n -tuple by an up-step and adds a duplicate of the first extended path as the last component. The original tuple is recovered from the first n components. For $n \geq 1$, a tuple of $(n + 1)$ -step paths ending in a down-step exists and is outside the image, proving strictness. Also $a_0 = a_1 = 1$.

The two model roots provide candidate thresholds on their respective parity lattices. Rounding a model root upward to the matching parity is correct whenever its distance from the relevant integer boundary exceeds the root error bound. Near a boundary, an exact comparison is needed. Thus (8.3) is a subinteger accuracy statement for matched continuous models, *not* an unconditional exact rounding theorem for all real y .

For the multiset sequence the large logarithmic scale becomes

$$g_b(x) = \lambda x^2 - 2x \log x + (c + 1)x - \frac{1}{2} \log x - \frac{1}{2} \log(2\pi). \quad (8.7)$$

One combines it with the same d_0, d_1 , the reciprocal-Stirling corrections, and, when required, the logarithm of the exponential-sector factor in (7.5). Its derivative is again $2\lambda x + O(\log x)$, so the same branchwise inversion and error argument applies. An exponential relative correction of order $x^3 2^{-x}$ changes the inverse by order $x^2 2^{-x}$, smaller than every algebraic inverse term.

9 Computational verification and reproducibility

9.1 What was checked exactly

The companion program `code/verify.py` separates exact integer assertions, symbolic algebra, and high-precision numerical calculations. All reported checks completed successfully. An independent dynamic program for nonnegative paths was compared with the reflection formula and the shifted binomial formula through $n = 100$: this tests 2601 ballot entries. The first 15 entries of each target sequence were compared with the OEIS data [1, 2, 5, 6]. The rising-factorial identity was tested exactly through $n = 25$; the fixed-power identities for $k = 1, 2$ through $n = 100$; and the adjacent-entry ratio through $n = 100$. Exact tied maxima were checked for every $n = s^2 - 2$ with $2 \leq s \leq 20$.

The coefficient generator uses rational symbolic arithmetic for (3.3), (3.7), and (3.10). It produces $R_0, \dots, R_6$, verifies their parities, and checks the first displayed polynomial formulas. These finite checks can catch an index shift or transcription error, but do not establish a uniform asymptotic theorem. The Gaussian bounds in Section 3 are the mathematical justification of that theorem.

9.2 Accuracy of the all-orders approximation

For selected indices, the numerical reference is the full sum over *all* endpoints, using exact integer ballot entries as bases and 75-decimal-digit logarithmic evaluation. The approximate value is the truncated expansion in Theorem 2.2. The signed error below is approximation divided by numerical reference, minus one. These are high-precision experiments, not interval-arithmetic certificates.

At $n = 1200$ the order-6 approximation has relative error about $7.52 \cdot 10^{-12}$. At square-adjacent indices some odd coefficient functions are small, and errors can be unusually favorable. No monotone improvement in J for every individual n is asserted. An asymptotic expansion controls fixed orders as n grows, not arbitrary truncation at a fixed small index.

n	$J = 0$	$J = 2$	$J = 4$	$J = 6$
25	$-5.892 \cdot 10^{-2}$	$-1.503 \cdot 10^{-4}$	$2.293 \cdot 10^{-5}$	$2.462 \cdot 10^{-6}$
50	$-6.268 \cdot 10^{-3}$	$-5.377 \cdot 10^{-4}$	$-1.202 \cdot 10^{-5}$	$-3.373 \cdot 10^{-7}$
100	$-1.608 \cdot 10^{-2}$	$-1.198 \cdot 10^{-5}$	$3.781 \cdot 10^{-7}$	$1.145 \cdot 10^{-8}$
200	$-1.075 \cdot 10^{-2}$	$6.415 \cdot 10^{-5}$	$4.041 \cdot 10^{-7}$	$2.464 \cdot 10^{-9}$
400	$-4.115 \cdot 10^{-3}$	$-7.936 \cdot 10^{-7}$	$5.982 \cdot 10^{-9}$	$4.673 \cdot 10^{-11}$
800	$-4.439 \cdot 10^{-3}$	$1.415 \cdot 10^{-5}$	$2.023 \cdot 10^{-8}$	$2.850 \cdot 10^{-11}$
1200	$-2.722 \cdot 10^{-3}$	$6.828 \cdot 10^{-6}$	$6.878 \cdot 10^{-9}$	$7.520 \cdot 10^{-12}$

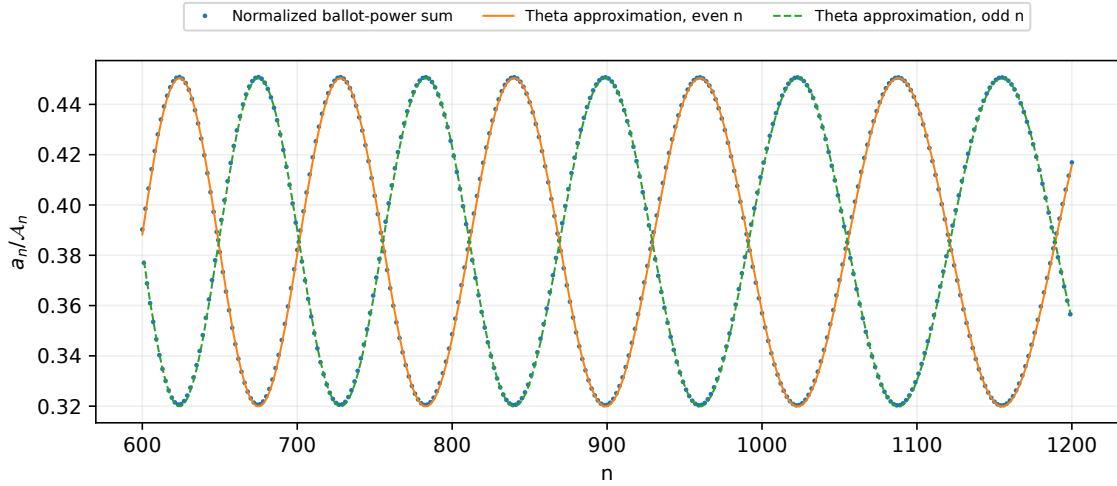
Table 1: Signed relative errors for truncations through order $m^{-J/2}$.

Figure 2: The normalized diagonal sequence and its leading approximation. Connecting the even and odd indices separately reveals the two smooth parity strands. The index-dependent widening of an oscillation is consistent with the square-root phase. Plotting computations use a localized log-Gamma sum; the full-sum tests are reported separately.

9.3 Exponential corrections, near-square transitions, and inverses

The first exponential sector was checked using exact integers for a_n , b_n , and $n!$. For $n = 20, 30, 40, 60, 80, 100$, the ratio of the exact relative correction to $\sqrt{\pi e} n^{3/2} 2^{-n} / (4\sqrt{2})$ ranges from approximately 0.99969 to 1.01631. Its order- $1/n$ phase dependence, rather than a single monotone convergence curve, is predicted by (7.5). The complete values are in `data/exponential_sector.csv`.

For the second localization transition, the code evaluates full sums at $s = 8, 16, 32$, $d = -2, -1, 0, 1, 2$, and $k = \text{round}(0.3s^3)$. The resulting upper-endpoint probabilities are compared with $(1 + \exp(-1.2d))^{-1}$ in `data/square_transition.csv`. The precise choice of the parity-preserving window $n = s^2 - 2 + 2d$ is tested rather than replaced by a nearby generic index.

The inverse computations use $L = \log a_n$ from the full endpoint sum. They compare (8.6) with a numerical root of the order-6 model of matching parity. Representative signed errors are:

n	Explicit inverse minus n	Order-6 model root minus n
100	$-9.714 \cdot 10^{-4}$	$-8.609 \cdot 10^{-11}$
200	$-3.112 \cdot 10^{-4}$	$-9.096 \cdot 10^{-12}$
400	$-9.734 \cdot 10^{-5}$	$-8.535 \cdot 10^{-14}$
800	$-2.994 \cdot 10^{-5}$	$-2.588 \cdot 10^{-14}$
1200	$-1.506 \cdot 10^{-5}$	$-4.543 \cdot 10^{-15}$

These measurements illustrate the extra factor of order $1/n$ gained when a logarithmic approximation is inverted. They do not provide a certified least threshold for the big- O statements.

9.4 Reproducing the files

From the extracted archive, run

```
python -m pip install -r requirements.txt
python code/verify.py --all
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The code uses Python 3.10 or newer with SymPy, mpmath, and matplotlib. The verification itself does not require network access; the OEIS values used in exact comparisons are embedded with their provenance. The first command is only for installing missing dependencies. Alternatively, `build.sh` compiles the article when the dependencies are already installed. The archive includes the figures used by the TeX source, raw CSV tables, generated coefficient polynomials, and the actual verification-status JSON.

No Lean file is included, because no formal proof was compiled for this report. No interval certificate is implied by a large working precision. The existing repo was inspected but not modified.

[Write note, batch 73O1.] In the collection the requirements file is `data/requirements.txt` and the build script is `code/build.sh`. The script changes into its own directory before calling `code/verify.py`, so it runs only in the delivered layout. `code/verify.py --all` rewrites `data/` and `figures/` in place, so the shipped records should be regenerated on a copy; the README gives commands that do this.

10 Further research questions and concrete next steps

The following questions are distinguished from the theorems already proved. They suggest a program extending the results rather than merely repeating the missing-constant question.

- Q1. Effective remainders and certified OEIS additions.** Make the constants in (3.11) explicit at orders $J = 0, 2, 4$, and combine them with interval evaluation of the theta sums. This would yield rigorous numerical enclosures for every sufficiently large index, an explicit log-convexity threshold, and certified threshold inversion. A finite exact check could then bridge the remaining initial range. The global trigamma estimate and the explicit two-site bound already provide useful non-asymptotic ingredients.
- Q2. Global strict log-convexity.** Do $a_{n-1}a_{n+1} > a_n^2$ and $b_{n-1}b_{n+1} > b_n^2$ hold at every $n \geq 1$? The limiting Turán gap proves eventual strictness, but does not settle the finite range

without explicit error constants. A direct injection, total-positivity argument, or inequality linking adjacent ballot rows might prove the global statement without enumeration.

- Q3. Uniform secondary transition expansions.** The exact two-site formula resolves the entire supercritical regime, and Corollary 6.4 resolves bounded square-window shifts. Develop uniform expansions when the integer displacement d grows with s , or when k/s^3 tends to zero or infinity simultaneously with d . The controlling variable is the exact $k \log \rho$, so a meaningful goal is a hierarchy of elementary scaled formulas with explicit error domains, not replacing an exact expression by an uncontrolled Taylor truncation.
- Q4. Optimal truncation and additional exponential scales.** The all-orders diagonal expansion is not an optimal-truncation theorem. Determine the large-order growth of the polynomials and theta-derivative coefficients, and whether a rigorously optimally truncated remainder can be decomposed into further exponential contributions. This problem is separate from the rigorously established tuple-multiset sectors; it should not be called resurgence without additional analysis.
- Q5. A growing number of multiset sectors.** Theorem 7.1 keeps Q fixed. Find useful ranges of $Q = Q(n)$ with uniform coefficient and remainder control, and study multisets whose size differs from the path length. The elementary symmetric coefficients of $0, \dots, n-1$ and the ratio k/m then interact in a two-parameter problem. An exact rising-factorial bridge remains available, but uniform asymptotic division requires new estimates.
- Q6. A reusable lattice-peak theorem for other path models.** Formulate sufficient hypotheses on a positive triangular array $C_{n,h}$ for a high-power sum to have a phase-dependent theta expansion. The proof here suggests the required data: a uniformly concave local logarithm, a growing interior peak, a fixed endpoint lattice, derivative expansions, and global tail domination. Apply such a theorem to weighted meanders and finite-step walks, then to multidimensional endpoint lattices where theta functions are associated with quadratic forms. This is the clearest route from the present OEIS case to a broad tool.

[Write note, batch 77P2, 2 October 2026.] The collection's report a380274-mahonian-growing-powers, in the same directory as this one (batch 77, manuscript 55), proves one more instance of the phenomenon for another positive triangular array, the Mahonian numbers $M_n(k)$ (permutations of n with k inversions; OEIS A380274, A380275 are $\sum_k M_n(k)^3$ and $\sum_k M_n(k)^4$). With V_n the inversion variance and C_n the central coefficient, its theorem "Complete growing-power leading behavior" gives $\sum_k M_n(k)^r / C_n^r \sim \Theta_\delta(r/V_n)$ for every $r = r_n \rightarrow \infty$, a discrete Gaussian theta function whose lattice shift $\delta \in \{0, \frac{1}{2}\}$ is fixed by $n \bmod 4$; there is no varying phase as in Theorem 6.2. Its proof uses Fourier estimates and log-concavity of the Mahonian row, not hypotheses of the kind asked for here. It is an instance, not the general theorem, and this question stays open. That report's section "Literature boundary and interpretation" says the same in a note and gives the dictionary between the two theta normalizations: its $Z_\delta(q) = \sum_{x \in \mathbb{Z} + \delta} e^{-qx^2/2}$ equals $\Theta_{q/8}(\delta)$ in the notation of (2.6), and its $\Theta_\delta(q) = e^{q\delta^2/2} Z_\delta(q)$; its subscript δ is a lattice shift, the subscript τ here a scale. Neither manuscript cites the other.

[Write note, batch 98D, 5 October 2026.] The question is answered in a sufficient-hypothesis form by Part II. Its Theorem 15.1 gives the theta limit for a general positive array under a local quadratic law, a Gaussian majorant and interior completeness, with arbitrary drifting phase; Theorem 17.2 gives every fixed order through a Bell-polynomial recurrence, and Theorem 18.1 the multidimensional version with a quadratic-form theta function. This report's critical regime satisfies the hypotheses (Proposition 20.1), and so does the Mahonian report's critical window once the Gaussian majorant is

weakened to an exponential one, which log-concavity supplies (Proposition 21.1). The sentence above that “this question stays open” is therefore superseded. Still open are a characterization, degenerate, boundary and multiple peaks, and concrete weighted-meander or multidimensional applications (Section 25). In Part II’s notation the Mahonian $Z_\delta(q)$ and this report’s $\Theta_{q/8}(\delta)$ are both $\vartheta(q, \delta)$ (equation (12.1)).

- Q7. Joint phase laws and higher discrete differences.** Determine effective joint laws of several neighboring normalized terms, and use them to analyze higher logarithmic differences or Jensen polynomials. Adjacent terms undergo a half-period shift rather than an independent random phase. The second difference is positive because its explicit theta quotient has a gap; analogous sign or hyperbolicity claims at higher orders require separate proofs.
- Q8. Sharper averaging errors.** The elementary discrepancy proof gives $O(\sqrt{N})$ counting error. For smooth periodic test functions, exploit the square-root phase, parity cancellation, and vanishing odd corrections to seek sharper Cesàro expansions. Such expansions may contain boundary-phase terms rather than a universal constant at the next order. An unsmoothed interval discrepancy and a smooth weighted average need not have the same best error term.
- Q9. The separate arithmetic conjecture.** The OEIS records Bala’s conjecture $a_{2p-1} \equiv 1 \pmod{p^3}$ for primes $p \geq 5$ [1]. Nothing in the present asymptotic proof establishes it. The exact shifted ballot formula gives a possible starting point for a prime-adic analysis, but real asymptotic errors, even exponentially small relative errors, have no direct force modulo p^3 . A future arithmetic report should maintain that distinction explicitly.
- Q10. Formalization with a visible theorem boundary.** Start with exact path semantics, reflection, shifted binomial identities, the adjacent-entry ratio, and the rising-factorial bridge. Next formalize the discrete two-site tail bound, which avoids Stirling theory. Only then build the analytic layer: gamma remainders, summable Gaussian bounds, phase-uniform termwise expansion, and theta identities. This staged plan could produce useful Lean theorems before the entire analytic result is available, without labeling the unformalized remainder as checked.

11 Conclusion

The central obstacle was not the exponential scale of the counts. That scale was already visible in the OEIS. The missing information lay in the interaction between a moving ballot peak and an endpoint lattice that remains coarse after taking a power proportional to the row index. Keeping that lattice produces an explicit theta multiplier, not a constant, and supports a complete algebraic expansion with uniform remainders.

The same analysis produces exact fluctuation envelopes and phase laws, strict eventual log-convexity, and stable branchwise inverse formulas. An exact adjacent-entry ratio further identifies the mode and controls all supercritical powers, including a second near-square transition. Finally, an exact symmetric-polynomial identity places the multiset sequence into a hierarchy of exponentially small corrections to the ordered-tuple sequence. Together these results turn a recorded nonconvergence phenomenon into a coherent analytic and combinatorial structure, while leaving the distinct arithmetic and formalization questions clearly identified.

Part II

A lattice-peak transfer theorem for growing power sums

12 Provenance, scope and notation of Part II

[Write note, batch 98D, 5 October 2026.] Part II was added in the write of batch 98D. It prints one manuscript, which answers question Q6 of Part I (Section 10) in a sufficient-hypothesis form: a general theorem of which Part I’s critical theta regime (Theorem 6.2) and the Mahonian report’s critical window are instances.

12.1 Provenance

The source is the archive `ProveIt_OEIS_Lattice_Peak_Transfer_2026-10-05.zip`, a late arrival of batch 98 of ProveIt’s incoming-reports intake (its twelfth manuscript, the only one of placement 98D), which arrived in commit `1ec443bc4` and was placed in this report by commit `3b9458b31`. It holds only the article `ProveIt_OEIS_Lattice_Peak_Transfer.tex` (641 lines) and its 14-page PDF, titled *A Lattice-Peak Transfer Theorem for Growing Power Sums*, with the subtitles “Phase-dependent theta asymptotics, all-orders corrections, multidimensional peaks, and inverse transfer” and “Applications to the OEIS ballot-power and Mahonian-power families”. It is dated 5 October 2026; its author line is “Research report for the ProveIt project”. Whether it was prepared with an AI assistant is not stated. It names no pinned commit: its source audit (Appendix D) says the repository facts were “checked against the ProveIt default branch on 5 October 2026”, and it knows the batch-77P2 note of 2 October 2026 at the end of question Q6 (commit `67d54b7ac`). Both reports it cites, this one and `a380274-mahonian-growing-powers`, were unchanged between that commit and the placement. No code, data, README or checksum list was delivered, so nothing was staged; the article and its PDF survive in the arrival commit. Part II is unrefereed, and nothing in it is formalized.

12.2 What the write changed

Section k of the manuscript is Section $k + 12$ here, and its Appendices A and B are Appendices C and D. Within each section every numbered statement and equation keeps the manuscript’s number after that shift (manuscript Theorem 3.1 is Theorem 15.1, equation (5.9) is (17.9)); the write’s additions follow the manuscript’s numbered items of their section. All labels carry the prefix `tbs:lpt:`. Symbols that collide with Part I or with the Mahonian report are renamed (Table 2); otherwise the text is the manuscript’s, with references to “the ballot report” turned into references to Part I. The write added, marked *[write]*:

- the correction of a false identity of the manuscript’s Section 5.1 (Remark 17.4, with a counterexample);
- the completion of two proofs: the theta-side tail in the all-orders theorem (Theorem 17.2) and the existence and localisation step of the inverse theorem (Theorem 19.1); a definition of “local completeness” in Theorem 18.1; a proof of Corollary 19.2;
- an exponential-majorant form of the transfer theorem (Theorem 15.4) and a log-concavity

criterion for it (Proposition 15.5), with which the Mahonian critical window is proved to fit the hypotheses (Proposition 21.1); the manuscript only predicted this;

- a verification that the ballot array satisfies the hypotheses (Proposition 20.1), the correction of the manuscript's claim that Part I's all-orders coefficients are "precisely" its own (Remark 20.2), the normalization of the cluster set (20.4), and the branchwise application of the inverse theorem to Part I's models (Remark 20.3);
- the section of further questions (Section 25), which keeps the manuscript's R1–R12 and adds every claim of the manuscript that is not proved.

Under Vladimir's standing rule of 4 October 2026 on unproved and wrong claims: one identity is refuted with a counterexample (Remark 17.4); one identification is shown overstated (Remark 20.2); the unproved claims are listed in Section 25, items F1–F7.

12.3 Notation

Three theta functions appear in this report and its neighbour, and they are normalized differently. Part II writes

$$\vartheta(q, \delta) = \sum_{j \in \mathbb{Z}} e^{-\frac{q}{2}(j-\delta)^2} = \Theta_{q/8}(\delta) = Z_\delta(q) = e^{-q\delta^2/2} \Theta_\delta^{\text{mgp}}(q), \quad (12.1)$$

where $\Theta_\tau(\alpha)$ is Part I's (2.6), whose subscript is a scale, and $Z_\delta(q)$ and $\Theta_\delta^{\text{mgp}}(q) = e^{q\delta^2/2} Z_\delta(q)$ are the Mahonian report's (its equation `mgp: eq: theta`), whose subscript is a lattice shift [17]. In particular Part I's $\Theta(\alpha) = \Theta_1(\alpha)$ is $\vartheta(8, \alpha)$, not $\vartheta(1, \alpha)$, and the Mahonian $\Theta_\delta^{\text{mgp}}(q)$ equals $\vartheta(q, \delta)$ only at $\delta = 0$. The identities (12.1) were checked numerically to 18 digits at three points.

Table 2: Symbols of Part II that differ from the manuscript or collide with Part I or the Mahonian report (mgp).

Part II	Meaning	Manuscript	Elsewhere
$\vartheta(q, \delta)$	translated theta series (14.3)	$\Theta(q, \delta)$	Part I Θ_τ , Θ ; mgp Z_δ , Θ_δ ; see (12.1)
$\vartheta_Q(\delta), Q_n, Q$	quadratic-form theta series and forms	Θ_Q, Q_n, Q	Part I Q : number of multiset sectors (Theorem 7.1)
$Q_m(u; q)$	Bell polynomials (17.4)	Q_m	as above
$\mathcal{T}_m(q, \delta)$	all-orders coefficients	same	Part I $T_\pm$: Turán constants
$L_n = \xi_n + d_n \mathbb{Z}, v_n$	lattice, offset, integer part	$\Lambda_n = a_n + d_n \mathbb{Z}, m_n$	Part I a_n (A357825), $\Lambda(\alpha)$, $m = n + 1$
$L_n = \xi_n + \mathbf{D}_n \mathbb{Z}^d$	d -dimensional lattice	$b_n + D_n \mathbb{Z}^d$	Part I b_n (A357871), $D_j(u)$
Ω_n	carrier amplitude	$A_n, A_n^{(0)}$	Part I $A_\ell(x)$, $\mathcal{A}_n$
t_n	small parameter of the all-orders theorem	ε_n	Part I $\varepsilon_n = m \bmod 2$; Part I $t = m^{-1/2}$ is t_n in Section 20
B_*, c_*	majorant constants of (H2)	B, c	Part I $B(n, h)$, c in (5.1), C, c in Lemma 3.2
$\mathcal{D}$	set of phase accumulation points	D	Part I $D_j(u)$
$\text{gap}(\delta)$	squared-distance gap in Proposition 16.1	$\gamma(\delta)$	—

Part II	Meaning	Manuscript	Elsewhere
$\omega_{n,j}, \gamma_{n,j}$	powered summand and Gaussian in proofs	$T_{n,j}, G_{n,j}$	Part I $T_{\pm}, g$
$\tilde{H}(x)$	non-theta part of H in Theorem 19.1	$h_0(x)$	Part I h : height
$I_n(k), z$	Mahonian numbers, q -factorial variable	$M_n(j), q$	Part I $M_p, M_{p,\tau}$: moments; mgp $M_n(k)$
σ_n^2 (Mahonian)	inversion variance	V_n	Part I V_m ; mgp V_n
r_n	the growing power	r	Part I: the power is k and $r = h + 1$; Section 20 uses Part I's letters
σ_n	curvature scale	same	Part I $\sigma \in \{0, 1\}$: parity branch (Section 8)
$R, R_n; R(x)$	window radii; remainder in (19.1)	same	Part I $R_j(u)$: correction polynomials
$K = [q_-, q_+]$	compact scale window	same	Part I K_n
$C_{n,x}; C_n$	array entry; mgp's central coefficient	same	Part I $C_{\pm}, C$
$\delta_n \in [-\frac{1}{2}, \frac{1}{2})$	lattice phase	same	mgp $\delta = \mu - \lfloor \mu \rfloor \in \{0, \frac{1}{2}\}$; ϑ is even and 1-periodic in δ , so the two conventions give the same values

13 Problem, scope, and relation to the OEIS reports

Let $C_{n,k} > 0$ be a positive triangular array, with k belonging to a one-dimensional lattice that may translate with n . For an exponent $r = r_n > 0$ consider

$$S_n(r) = \sum_k C_{n,k}^r. \quad (13.1)$$

When r is fixed and the row has a central-limit shape, the usual local-limit/Riemann-sum heuristic gives a Gaussian integral. When r grows, raising the row to the r th power narrows its effective width by $r^{-1/2}$. The transition occurs when that width becomes comparable to the lattice spacing. At that point a Riemann integral is no longer the correct leading object: the individual lattice sites survive, and the answer is a theta sum.

This mechanism is visible in two existing ProveIt reports. The report on the ballot array $S_{n,k} = \sum_h B(n, h)^k$ (Part I) proves a critical theta regime for $k \asymp n$ and asks, as its Question Q6, for “a reusable lattice-peak theorem for other path models.” A later report on Mahonian power sums proves the same type of transition when the power is comparable to the inversion variance, but explicitly records itself as an instance rather than a solution of Q6. The public OEIS entry A357825 records the diagonal ballot power sum and a nonconvergent normalized ratio, while A380275 records fixed-power Mahonian asymptotics; Wang’s 2026 preprint proves the fixed-real-power Mahonian expansion but does not claim uniformity in a growing power [1, 14, 15].

The present report addresses precisely the missing abstraction. It deliberately does *not* try to supersede general discrete Laplace theory. Recent work of Helton, Hughes and Schlosser [8] treats discrete Laplace asymptotics in the regime where a lattice becomes fine in the natural

Laplace scale. Our focus is the complementary critical regime in which the rescaled lattice has a nonzero limiting mesh, so the lattice survives as a theta multiplier.

Main contributions

The new statements proved here are:

- (i) a one-dimensional lattice-peak transfer theorem with arbitrary drifting phase and uniformity on compact critical windows;
- (ii) a stable version for subsequences with only compactness, yielding the complete set of theta cluster values;
- (iii) an all-orders theorem whose coefficients are generated by a universal Bell-polynomial recurrence and can be written through theta derivatives;
- (iv) a multidimensional version with a positive-definite Hessian and a translated lattice theta function;
- (v) an inverse-transfer theorem showing exactly how the theta factor perturbs a smooth inverse;
- (vi) a dictionary recovering the leading critical theorems of the ballot and Mahonian OEIS reports as instances.

The theorem is sufficient, not necessary: many arrays will satisfy weaker hypotheses. Its value is that each hypothesis is checkable by standard local asymptotic and tail tools.

[write] For item (vi): the ballot array satisfies the hypotheses (Proposition 20.1); the Mahonian array satisfies them with the Gaussian majorant (H2) replaced by the exponential majorant (H2e) of Theorem 15.4 (Proposition 21.1). The manuscript's abstract says that both "fit the hypotheses"; its Section 9 only predicts it for the Mahonian array.

14 The lattice-peak normalization

Definition 14.1 (Moving lattice and phase). For each n , let

$$L_n = \xi_n + d_n \mathbb{Z}, \quad d_n > 0,$$

and let $x_n \in \mathbb{R}$ be a distinguished continuous peak location. Choose $\delta_n \in [-1/2, 1/2)$ so that

$$x_n = \xi_n + d_n(\nu_n + \delta_n) \tag{14.1}$$

for some $\nu_n \in \mathbb{Z}$. Thus the lattice sites near the peak are

$$x_{n,j} = x_n + d_n(j - \delta_n), \quad j \in \mathbb{Z}.$$

The number δ_n is the *lattice phase* of the continuous peak.

Let $\Omega_n > 0$ be a continuous carrier amplitude and $\sigma_n \rightarrow \infty$ the curvature scale. We write

$$q_n = \frac{r_n d_n^2}{\sigma_n^2}. \tag{14.2}$$

The critical window is $q_n \asymp 1$. The basic translated theta function is

$$\vartheta(q, \delta) = \sum_{j \in \mathbb{Z}} \exp\left[-\frac{q}{2}(j - \delta)^2\right], \quad q > 0, \quad (14.3)$$

which is 1-periodic and even in δ modulo integers.

Definition 14.2 (Critical lattice-peak hypotheses). Fix a compact interval $K = [q_-, q_+] \subset (0, \infty)$. A family $C_{n,k}$ satisfies the critical lattice-peak hypotheses on K if, whenever $r_n > 0$ is chosen so that $q_n \in K$, the following hold.

(H1) **Local quadratic law.** For every fixed $R > 0$,

$$\sup_{|j - \delta_n| \leq R} \left| r_n \log \frac{C_{n,x_{n,j}}}{\Omega_n} + \frac{q_n}{2}(j - \delta_n)^2 \right| \rightarrow 0. \quad (14.4)$$

Here $C_{n,x}$ denotes the row entry at the lattice site x .

(H2) **Uniform Gaussian domination.** There are constants $B_*, c_* > 0$, independent of n and of $q_n \in K$, such that for every admissible lattice site,

$$\left(\frac{C_{n,x_{n,j}}}{\Omega_n} \right)^{r_n} \leq B_* \exp[-c_*(j - \delta_n)^2]. \quad (14.5)$$

(H3) **Interior completeness.** For every fixed R , all sites with $|j - \delta_n| \leq R$ are admissible for sufficiently large n .

The hypotheses separate the proof into the two ingredients that repeatedly appear in concrete OEIS analyses: a local Taylor/Stirling/Edgeworth expansion and a global log-concavity or Fourier tail estimate. Notice that no probabilistic interpretation is required.

15 The critical theta transfer theorem

Theorem 15.1 (One-dimensional lattice-peak transfer). *Assume the hypotheses of Definition 14.2. Then, uniformly for all exponent choices with $q_n \in K$,*

$$\frac{S_n(r_n)}{\Omega_n^{r_n}} = \vartheta(q_n, \delta_n) + o(1). \quad (15.1)$$

More generally, if the local law has a uniform additive error $\eta_n(R)$ on $|j - \delta_n| \leq R$, then for every fixed R

$$\begin{aligned} \left| \frac{S_n(r_n)}{\Omega_n^{r_n}} - \vartheta(q_n, \delta_n) \right| &\leq \left(e^{\eta_n(R)} - 1 \right) \sum_{|j - \delta_n| \leq R} e^{-q_n(j - \delta_n)^2/2} \\ &\quad + B_* \sum_{|j - \delta_n| > R} e^{-c_*(j - \delta_n)^2} + \sum_{|j - \delta_n| > R} e^{-q_n(j - \delta_n)^2/2}, \end{aligned} \quad (15.2)$$

for all sufficiently large n .

Proof. Write

$$\omega_{n,j} = \left(C_{n,x_{n,j}} / \Omega_n \right)^{r_n}, \quad \gamma_{n,j} = \exp[-q_n(j - \delta_n)^2/2].$$

Fix R . By interior completeness all lattice sites in the finite window $|j - \delta_n| \leq R$ occur in the row for large n . The local quadratic law gives

$$\omega_{n,j} = \gamma_{n,j} e^{e_{n,j}}, \quad |e_{n,j}| \leq \eta_n(R),$$

there, proving the first term in (15.2). Outside the window, hypothesis (H2) bounds the actual row tail by the second term, while $q_n \geq q_-$ bounds the omitted theta tail by the third. The two Gaussian tails tend to zero uniformly as $R \rightarrow \infty$, and then $\eta_n(R) \rightarrow 0$ for fixed R . This proves (15.1) by the standard “first n , then R ” argument. Every estimate is uniform in $q_n \in K$ and in the phase because translating a unit lattice does not change the elementary Gaussian tail bound by more than an absolute constant. $\square$

Corollary 15.2 (Critical subsequential limits and phase law). *Suppose $q_n \rightarrow q \in (0, \infty)$. If $\delta_{n_j} \rightarrow \delta$ along a subsequence, then*

$$\frac{S_{n_j}(r_{n_j})}{\Omega_{n_j}^{r_{n_j}}} \rightarrow \vartheta(q, \delta). \quad (15.3)$$

Consequently, if the set of phase accumulation points is $\mathcal{D} \subset \mathbb{R}/\mathbb{Z}$, the set of accumulation points of the normalized power sum is exactly

$$\vartheta(q, \mathcal{D}) = \{\vartheta(q, \delta) : \delta \in \mathcal{D}\}. \quad (15.4)$$

If the phases are dense modulo 1, the cluster set is the full compact interval

$$[\min_{\delta} \vartheta(q, \delta), \max_{\delta} \vartheta(q, \delta)].$$

Proof. The theta series and all its phase derivatives converge locally uniformly for $q > 0$, so $\vartheta(q_n, \delta_n) \rightarrow \vartheta(q, \delta)$ on any convergent phase subsequence. Theorem 15.1 supplies the same limit for the normalized power sum. Compactness of the circle gives the converse inclusion for cluster points. Continuity maps a connected dense phase closure to the interval between its extrema. $\square$

Remark 15.3 (Why a “constant amplitude” can fail). At fixed power, a local central limit theorem often produces one Gaussian integral and hence one constant. In the critical regime, the mesh after peak rescaling is asymptotically nonzero. The phase δ_n is therefore not an error term: it is part of the leading order. Replacing $\vartheta(q, \delta_n)$ by its phase average is generally false pointwise, even when it is correct after Cesàro averaging under an equidistribution hypothesis.

[write] The proof of Theorem 15.1 uses (H2) only through the second term of (15.2), that is, through a phase-uniformly summable majorant. The Gaussian form of (H2) can therefore be weakened, and the weakened form follows from log-concavity. The uniformity “for all exponent choices” in Theorem 15.1 follows from the statement for single sequences by the usual argument: a sequence of counterexamples at indices $n_k \rightarrow \infty$ can be completed, by any exponents with $q_n \in K$ at the other indices, to a sequence violating the single-sequence statement.

Theorem 15.4 (Exponential-majorant transfer; [write]). *Let r_n be a sequence of exponents with $q_n \in K$ for which (H1) and (H3) hold, and suppose instead of (H2):*

(H2e) *there are $B_*, c_* > 0$ and n_0 such that for $n \geq n_0$ and every admissible site*

$$\left(\frac{C_{n,x_{n,j}}}{\Omega_n} \right)^{r_n} \leq B_* \exp[-c_* |j - \delta_n|]. \quad (15.5)$$

Then $S_n(r_n)/\Omega_n^{r_n} = \vartheta(q_n, \delta_n) + o(1)$, and (15.2) holds for $n \geq n_0$ with its second term replaced by $B_* \sum_{|j-\delta_n|>R} e^{-c_*|j-\delta_n|}$.

Proof. Repeat the proof of Theorem 15.1. The only change is the bound for the actual row tail. For $R \geq 1$ and $\delta \in [-\frac{1}{2}, \frac{1}{2})$, the sites with $|j - \delta| > R$ have $|j| > R - \frac{1}{2}$, so

$$\sum_{|j-\delta|>R} e^{-c_*|j-\delta|} \leq 2 \sum_{i>R-1/2} e^{-c_*(i-1/2)} \leq \frac{2e^{-c_*(R-1)}}{1 - e^{-c_*}},$$

which tends to 0 as $R \rightarrow \infty$ uniformly in the phase. The window term and the theta tail are unchanged. $\square$

Proposition 15.5 (Log-concave rows; [write]). *Suppose that for all large n the admissible sites form a set of consecutive j , on which $j \mapsto C_{n,x_{n,j}}$ is positive and log-concave: $C_{n,x_{n,j}}^2 \geq C_{n,x_{n,j-1}} C_{n,x_{n,j+1}}$. Let r_n have $q_n \in K$ and satisfy (H1) and (H3). Then (H2e) holds with*

$$B_* = e^{1+3q_-}, \quad c_* = q_-.$$

Consequently $S_n(r_n)/\Omega_n^{r_n} = \vartheta(q_n, \delta_n) + o(1)$.

Proof. Write $u_j = j - \delta_n$ and $\ell_j = \log \omega_{n,j}$, a concave sequence on the admissible set (because $r_n > 0$). By (H1) and (H3) with $R = 3$, for large n every site with $|u_j| \leq 3$ is admissible and $|\ell_j + q_n u_j^2/2| \leq \eta$ there, where $\eta = \eta_n(3) \leq \min(1, q_-/4)$.

Let j_1 be the least integer with $u_{j_1} \geq 1$, so $u_{j_1} \in [1, 2)$ and $u_{j_1+1} \in [2, 3)$. Then

$$\ell_{j_1+1} - \ell_{j_1} \leq -\frac{q_n}{2}(u_{j_1+1}^2 - u_{j_1}^2) + 2\eta = -\frac{q_n}{2}(2u_{j_1} + 1) + 2\eta \leq -\frac{3}{2}q_- + 2\eta \leq -q_-.$$

By concavity the increments $\ell_{j+1} - \ell_j$ do not increase, so for every admissible $j \geq j_1 + 1$,

$$\ell_j \leq \ell_{j_1+1} - q_-(j - j_1 - 1) \leq \eta - q_-(u_j - 3) \leq 1 + 3q_- - q_-|u_j|,$$

using $\ell_{j_1+1} \leq -2q_n + \eta \leq \eta$ and $u_{j_1+1} < 3$. The sites with $u_j \leq -2$ are treated in the same way from the greatest j_{-1} with $u_{j_{-1}} \leq -1$. The remaining sites have $|u_j| < 3$, where $\ell_j \leq \eta \leq 1 \leq 1 + 3q_- - q_-|u_j|$. This is (15.5). Theorem 15.4 gives the conclusion. $\square$

16 Subcritical and supercritical companions

The critical theorem is the part needed to answer Q6, but the same normalization clarifies both adjacent regimes.

Proposition 16.1 (Theta limits at small and large q). *Uniformly in $\delta \in \mathbb{R}$,*

$$\vartheta(q, \delta) = \sqrt{\frac{2\pi}{q}} \left(1 + O\left(e^{-2\pi^2/q}\right) \right), \quad q \downarrow 0, \quad (16.1)$$

$$\vartheta(q, \delta) = \sum_{j: |j-\delta|=d(\delta)} e^{-q d(\delta)^2/2} \left(1 + O\left(e^{-q \text{gap}(\delta)/2}\right) \right), \quad q \rightarrow \infty, \quad (16.2)$$

where $d(\delta) = \text{dist}(\delta, \mathbb{Z})$ and $\text{gap}(\delta)$ is the gap between the squared distance of a nearest lattice point and the next distinct squared distance. Away from half-integer phases, the first sum in (16.2) has one term; at half-integer phase it has two equal terms.

Proof. Poisson summation gives

$$\vartheta(q, \delta) = \sqrt{\frac{2\pi}{q}} \sum_{m \in \mathbb{Z}} e^{-2\pi^2 m^2 / q} e^{2\pi i m \delta},$$

from which (16.1) follows. The large- q statement is direct: isolate the nearest lattice point or tied nearest pair and bound the remaining geometric-Gaussian tail by the first omitted squared-distance gap. $\square$

[write] The Poisson form is Part I's (4.1) at $\tau = q/8$. In (16.2) the implied constant is uniform in δ , but the rate is not: $\text{gap}(\delta) \rightarrow 0$ as δ approaches a half-integer, where the bound degenerates until the tied pair is used. (Here $d(\delta)$ is the manuscript's $d(\delta)$, renamed only to avoid the lattice span d_n .)

Remark 16.2 (What extra hypotheses are needed outside the critical window). Theorem 15.1 assumes q_n remains in a fixed compact subset of $(0, \infty)$. Letting $q_n \rightarrow 0$ requires the local expansion on a window whose number of lattice sites grows like $q_n^{-1/2}$; letting $q_n \rightarrow \infty$ requires errors small enough after multiplication by q_n at the one or two nearest sites. These are genuinely stronger demands. In applications, the subcritical side is usually obtained from a uniform local limit theorem, while the far supercritical side is often better handled by exact adjacent-entry ratios, as in the ballot report (Theorem 6.3). [write: this remark is a heuristic; it is item F2 of Section 25.]

17 An all-orders transfer theorem

The leading theta theorem is only the first layer. Concrete OEIS problems often provide a full local logarithmic expansion, and then every coefficient can be transferred through the lattice sum.

Definition 17.1 (Polynomial local expansion). Let $t_n \rightarrow 0$. For a fixed integer $J \geq 0$, suppose that on windows $|j - \delta_n| \leq R_n$ with $R_n \rightarrow \infty$ slowly enough,

$$r_n \log \frac{C_{n,x_{n,j}}}{\Omega_n} = -\frac{q_n}{2} u^2 + \sum_{\ell=1}^J t_n^\ell P_\ell(u; q_n) + O\left(t_n^{J+1} (1 + |u|)^M\right), \quad u = j - \delta_n, \quad (17.1)$$

where each P_ℓ is a polynomial in u , with coefficients continuous in $q \in K$, and the remainder is uniform. Assume moreover that for some $c_*, B_* > 0$ and some exponent M' , uniformly in n and phase,

$$\left(\frac{C_{n,x_{n,j}}}{\Omega_n} \right)^{r_n} (1 + |j - \delta_n|)^{M'} \leq B_* e^{-c_*(j - \delta_n)^2}, \quad (17.2)$$

and that the chosen $R_n \rightarrow \infty$ satisfies

$$\sum_{|j - \delta_n| > R_n} (1 + |j - \delta_n|)^{M'} e^{-c_*(j - \delta_n)^2} = O(t_n^{J+1}) \quad (17.3)$$

uniformly in the phase. Enlarging M' if necessary is harmless.

Define polynomials Q_m recursively by

$$Q_0 = 1, \quad m Q_m = \sum_{\ell=1}^m \ell P_\ell Q_{m-\ell} \quad (m \geq 1). \quad (17.4)$$

Thus

$$Q_1 = P_1, \quad Q_2 = P_2 + \frac{1}{2}P_1^2, \quad Q_3 = P_3 + P_1P_2 + \frac{1}{6}P_1^3.$$

Theorem 17.2 (All-orders lattice transfer). *Under Definition 17.1, including (17.2)–(17.3),*

$$\frac{S_n(r_n)}{\Omega_n^{r_n}} = \sum_{m=0}^J t_n^m \mathcal{T}_m(q_n, \delta_n) + \mathcal{O}(t_n^{J+1}), \quad (17.5)$$

where

$$\mathcal{T}_m(q, \delta) = \sum_{j \in \mathbb{Z}} Q_m(j - \delta; q) e^{-q(j-\delta)^2/2}. \quad (17.6)$$

The error is uniform for $q_n \in K$ and all phases. If the coefficients of P_ℓ are smooth in q , then every $\mathcal{T}_m$ is smooth in (q, δ) .

Proof. Exponentiating (17.1) gives

$$\exp\left(\sum_{\ell=1}^J t_n^\ell P_\ell\right) = \sum_{m=0}^J t_n^m Q_m + \mathcal{O}\left(t_n^{J+1}(1+|u|)^{M'}\right)$$

for some M' , uniformly on the retained window. The coefficient recurrence is obtained by differentiating the formal identity $Q(t) = \exp(\sum_{\ell \geq 1} t^\ell P_\ell)$ and comparing coefficients in $tQ'(t) = Q(t) \sum_{\ell \geq 1} \ell t^{\ell-1} P_\ell$. Multiplication by the leading Gaussian and summation are justified termwise by (17.2). The complement of the local window is $\mathcal{O}(t_n^{J+1})$ by (17.3). Absolute and locally uniform convergence also permits differentiation in q and δ .

[write] *Completion.* The last-but-one sentence bounds the actual summands outside the window, $\omega_{n,j} \leq B_*(1+|u|)^{M'} e^{-c_* u^2}$; the proof also has to discard the *theta-side* tail $\sum_{|u| > R_n} Q_m(u; q_n) e^{-q_n u^2/2}$, $m \leq J$, which it does not mention. Assume $M' > 0$. Fix $U \geq 1$ with $M' \log(1+U) \geq \log B_* + 2$, and for each n a site with $|u| \in [U, U+1]$ (one exists for every phase). For large n this site lies in the window, where (17.1) gives $\omega_{n,j} \geq e^{-q_n u^2/2-1}$ (the polynomials P_ℓ have coefficients bounded on K , and $t_n \rightarrow 0$). Inserting this into (17.2) gives $(c_* - q_n/2)u^2 \leq \log B_* + 1 - M' \log(1+|u|) \leq -1$, hence

$$c_* \leq \frac{q_n}{2} - \kappa, \quad \kappa = (U+1)^{-2},$$

for all large n . Since $|Q_m(u; q)| \leq C_m(1+|u|)^{e_m}$ for $q \in K$,

$$\sum_{|u| > R_n} |Q_m(u; q_n)| e^{-q_n u^2/2} \leq C_m \sup_{v \in \mathbb{R}} \left[(1+|v|)^{e_m - M'} e^{-\kappa v^2} \right] \sum_{|u| > R_n} (1+|u|)^{M'} e^{-c_* u^2},$$

which is $\mathcal{O}(t_n^{J+1})$ by (17.3). With the two tails discarded, the window sum of the expansion completes (17.5). On the window, “ $R_n \rightarrow \infty$ slowly enough” is read as: the exponent $\sum_{\ell} t_n^\ell P_\ell + \mathcal{O}(t_n^{J+1}(1+|u|)^{M'})$ stays bounded for $|u| \leq R_n$, which is what the finite Taylor expansion of the exponential needs. The manuscript’s sentence “Enlarging M' if necessary is harmless” is not justified as stated: a larger M' in (17.2) forces a smaller c_* , and (17.3) with the smaller c_* is a stronger requirement on R_n . The completion above needs only $M' > 0$; the case $M' = 0$ is item F6 of Section 25. $\square$

17.1 Theta-derivative calculus

Polynomial-Gaussian sums in (17.6) can be reduced to theta derivatives. From

$$\partial_\delta e^{-qu^2/2} = que^{-qu^2/2}, \quad \partial_q e^{-qu^2/2} = -\frac{1}{2}u^2 e^{-qu^2/2},$$

one obtains, for example,

$$\sum_j u e^{-qu^2/2} = q^{-1} \partial_\delta \vartheta, \quad (17.7)$$

$$\sum_j u^2 e^{-qu^2/2} = -2 \partial_q \vartheta, \quad (17.8)$$

$$\sum_j u^3 e^{-qu^2/2} = q^{-1} \partial_\delta (-2 \partial_q \vartheta) + 2q^{-2} \partial_\delta \vartheta. \quad (17.9)$$

[write: (17.9) is corrected; the manuscript omits its last term. See Remark 17.4.] Hence every coefficient in Theorem 17.2 is a finite differential operator, with rational functions of q as coefficients, applied to $\vartheta(q, \delta)$. This is the abstract explanation for the theta-derivative structures that appear separately in the ballot and Mahonian calculations.

Corollary 17.3 (Parity cancellation). *Suppose $P_\ell(-u; q) = (-1)^\ell P_\ell(u; q)$. Then $Q_m(-u; q) = (-1)^m Q_m(u; q)$. At symmetric phases $\delta \equiv 0$ or $1/2 \pmod{1}$, every odd coefficient $\mathcal{T}_{2s+1}(q, \delta)$ vanishes.*

Proof. The parity statement follows inductively from (17.4). At integer or half-integer phase, the translated lattice is invariant under $u \mapsto -u$, so every odd polynomial-Gaussian sum cancels term by term. $\square$

Remark 17.4 (Correction of the third identity; [write]). The manuscript prints the third identity as

$$\sum_j u^3 e^{-qu^2/2} \stackrel{?}{=} q^{-1} \partial_\delta (-2 \partial_q \vartheta), \quad (17.10)$$

which is false. Since $u = j - \delta$, $\partial_\delta (u^2 e^{-qu^2/2}) = (-2u + qu^3) e^{-qu^2/2}$; summing and using (17.7),

$$q^{-1} \partial_\delta (-2 \partial_q \vartheta) = \sum_j u^3 e^{-qu^2/2} - \frac{2}{q} \sum_j u e^{-qu^2/2} = \sum_j u^3 e^{-qu^2/2} - 2q^{-2} \partial_\delta \vartheta,$$

which is (17.9). The dropped term is not zero in general. *Counterexample:* at $q = 2.7, \delta = 0.31$ the left side of (17.10) is $0.0189884 \dots (0.01898839563097749448 \dots)$, its right side is $0.0222562 \dots (0.02225616196228909469 \dots)$, and the right side of (17.9) agrees with the left to 40 digits (direct lattice sums, the derivatives taken termwise; the printed form was also evaluated by numerical differentiation, with the same value). The dropped term vanishes exactly when $\partial_\delta \vartheta = 0$, that is, at integer and half-integer phases, where both sides of (17.10) are zero.

The general claim after the identities is true, and in a sharper form: with $\Sigma_p = \sum_j u^p e^{-qu^2/2}$, the same differentiation gives $\partial_\delta \Sigma_p = -p \Sigma_{p-1} + q \Sigma_{p+1}$, so

$$\Sigma_0 = \vartheta, \quad \Sigma_{p+1} = q^{-1} \partial_\delta \Sigma_p + p q^{-1} \Sigma_{p-1} \quad (p \geq 0, \Sigma_{-1} = 0), \quad (17.11)$$

and every Σ_p is a polynomial in q^{-1} times phase derivatives of ϑ alone; for instance $\Sigma_2 = q^{-2} \partial_\delta^2 \vartheta + q^{-1} \vartheta$ and $\Sigma_3 = q^{-3} \partial_\delta^3 \vartheta + 3q^{-2} \partial_\delta \vartheta$. This is Part I's moment recurrence (4.2) in the other normalization: Part I's moments are $M_{p,\tau} = 2^p \Sigma_p$ at $q = 8\tau$, and at $q = 8$ the two formulas give Part I's $M_2 = \Theta''/16 + \Theta/2$ and $M_3 = \Theta'''/64 + 3\Theta'/8$ (Appendix A). At the point of the counterexample the formula for Σ_3 agrees with the direct sum to 40 digits.

18 A multidimensional lattice-peak theorem

Many combinatorial arrays have several coupled endpoint statistics. The same proof survives with almost no change.

Let $L_n = \xi_n + \mathbf{D}_n \mathbb{Z}^d$ be a full-rank lattice in $\mathbb{R}^d$, where $\mathbf{D}_n \rightarrow \mathbf{D}$ with $\mathbf{D}$ invertible after any desired normalization. Write the peak as $x_n = \xi_n + \mathbf{D}_n(v_n + \delta_n)$ with $\delta_n \in [-1/2, 1/2)^d$. Let H_n be a positive-definite curvature matrix and assume the powered local logarithm has the form

$$r_n \log \frac{C_{n,x_n,j}}{\Omega_n} = -\frac{1}{2}(j - \delta_n)^T \mathbf{Q}_n (j - \delta_n) + o(1), \quad (18.1)$$

for fixed j , where

$$\mathbf{Q}_n = r_n \mathbf{D}_n^T H_n \mathbf{D}_n \longrightarrow \mathbf{Q} > 0.$$

Define

$$\vartheta_{\mathbf{Q}}(\delta) = \sum_{j \in \mathbb{Z}^d} \exp \left[-\frac{1}{2}(j - \delta)^T \mathbf{Q} (j - \delta) \right]. \quad (18.2)$$

Theorem 18.1 (Multidimensional transfer). *Assume (18.1), local completeness, and a uniform summable bound*

$$\left(C_{n,x_n,j} / \Omega_n \right)^{r_n} \leq B_* \exp[-c_* \|j - \delta_n\|^2].$$

Then

$$\frac{S_n(r_n)}{\Omega_n^{r_n}} = \vartheta_{\mathbf{Q}_n}(\delta_n) + o(1). \quad (18.3)$$

If $\mathbf{Q}_n \rightarrow \mathbf{Q}$ and $\delta_n \rightarrow \delta$ along a subsequence, the right side tends to $\vartheta_{\mathbf{Q}}(\delta)$.

Proof. The proof of Theorem 15.1 applies verbatim with Euclidean balls replacing intervals. The Gaussian majorant is summable on $\mathbb{Z}^d$, and the family of tails is uniform because the phases remain in a compact fundamental cube and the eigenvalues of $\mathbf{Q}_n$ stay bounded above and below on a compact positive-definite set. $\square$

[write] “Local completeness” is not defined in the manuscript; it is read as the d -dimensional form of (H3): for every fixed R , all sites with $\|j - \delta_n\| \leq R$ are admissible for large n . The local law (18.1) is assumed for fixed j ; since the phases lie in a compact cube, it holds uniformly on every ball $\|j - \delta_n\| \leq R$, which is what the window step uses. The bounds on the eigenvalues of $\mathbf{Q}_n$ hold for large n because $\mathbf{Q}_n \rightarrow \mathbf{Q} > 0$.

Remark 18.2 (Root-lattice and constrained variants). Nothing essential requires the lattice to be a copy of $\mathbb{Z}^d$. A fixed full-rank lattice is reduced to this form by a basis matrix. If the peak lies on an affine sublattice or a linear conservation law, one first parametrizes that sublattice and applies the theorem in its intrinsic dimension. Boundary peaks require half-theta or cone-theta functions and are a separate extension. [write: the last two sentences are not proved here; item F3 and question R3 of Section 25.]

19 Inverse transfer: how theta oscillations move thresholds

Growing combinatorial sequences are often inverted: given a large target Y , one asks for the real or integer index at which the count reaches Y . A phase-dependent multiplier produces a systematic correction.

Theorem 19.1 (One-step inverse transfer). *Let $F(x)$ be a positive smooth interpolation of a growing sequence on an interval $x \rightarrow \infty$, and suppose*

$$\log F(x) = G(x) + H(x) + R(x), \quad (19.1)$$

where $G'(x) > 0$, $G'(x) \rightarrow \infty$ or stays bounded away from 0, $H(x) = O(1)$, and $R(x) = o(1)$. Let $x_0 = x_0(Y)$ solve $G(x_0) = \log Y$. Assume

$$\frac{G''(x_0)}{G'(x_0)^2} = o(1), \quad \frac{H'(x_0)}{G'(x_0)} = o(1), \quad \frac{R(x_0)}{1 + |H(x_0)|} = o(1). \quad (19.2)$$

Then the solution $x = x(Y)$ of $F(x) = Y$ satisfies

$$x = x_0 - \frac{H(x_0)}{G'(x_0)} + o\left(\frac{1 + |H(x_0)|}{G'(x_0)}\right). \quad (19.3)$$

If

$$H(x) = \log \vartheta(q(x), \delta(x)) + \tilde{H}(x),$$

then the theta factor contributes exactly

$$-\frac{\log \vartheta(q(x_0), \delta(x_0))}{G'(x_0)} \quad (19.4)$$

to first order.

Proof. Put $\Delta = x - x_0$. Taylor expansion gives

$$0 = G'(x_0)\Delta + H(x_0) + R(x_0) + O(G''(x_0)\Delta^2) + O(H'(x_0)\Delta) + o(|\Delta|G'(x_0)).$$

The candidate $\Delta_0 = -H(x_0)/G'(x_0)$ is $O((1 + |H|)/G')$. Substituting this scale and using (19.2) shows that every remaining term is $o(1 + |H|)$. Division by $G'(x_0)$ proves (19.3).

[write] *Completion.* The proof above presupposes a solution x with Δ of the order of the candidate, and evaluates G'' and H' at x_0 although Taylor's theorem needs them between x_0 and x . We read $G \in C^2$, $H \in C^1$, R continuous on some $[x_1, \infty)$, and the $o(1)$ in (19.2) as $Y \rightarrow \infty$. Because G is continuous and increases to ∞ (G' is bounded below), $x_0(Y)$ runs through every large real number, so (19.2) says $G''/G'^2 \rightarrow 0$ and $H'/G' \rightarrow 0$ as $x \rightarrow \infty$; and the third condition follows from $R = o(1)$.

Existence and localisation. Let $K_0 \geq \sup_{x \geq x_1} (|H| + |R|)$, enlarging x_1 if needed, and let $X = G^{-1}$. For $s \in I = [\log Y - K_0 - 1, \log Y + K_0 + 1]$, $\frac{d}{ds} \log G'(X(s)) = G''(X(s))/G'(X(s))^2 = o(1)$, so $G' = G'(x_0)(1 + o(1))$ on $X(I)$, and $X(I)$ lies within $2(K_0 + 1)/G'(x_0)$ of x_0 for large Y . At $x_{\pm} = X(\log Y \pm (K_0 + 1))$, $\log F(x_{\pm}) - \log Y = \pm(K_0 + 1) + H(x_{\pm}) + R(x_{\pm})$ is positive at x_+ and negative at x_- , so $F = Y$ has a root between them by the intermediate value theorem. Conversely every root $x \geq x_1$ has $|G(x) - \log Y| = |H(x) + R(x)| \leq K_0$, so it lies in $X(I)$. Uniqueness is not claimed; the conclusion holds for every root.

Taylor step. For a root x and $\Delta = x - x_0$, $|\Delta| \leq 2(K_0 + 1)/G'(x_0)$ and

$$0 = G'(x_0)\Delta + \frac{1}{2}G''(\xi)\Delta^2 + H(x_0) + H'(\zeta)\Delta + R(x)$$

with $\xi, \zeta \in X(I)$. There $G''(\xi) = o(G'(x_0)^2)$ and $H'(\zeta) = o(G'(x_0))$, so the second, fourth and fifth terms are $o(1)$. Hence $G'(x_0)\Delta = -H(x_0) + o(1)$, which implies (19.3) (with $o(1/G'(x_0))$, the same thing because H is bounded). The last assertion follows by inserting $H = \log \vartheta + \tilde{H}$. $\square$

Corollary 19.2 (No universal constant inverse correction under drifting phase). *If $q(x) \rightarrow q_* > 0$ while $\delta(x)$ has at least two accumulation points at which $\vartheta(q_*, \delta)$ takes different values, then no replacement of the theta term in (19.4) by a single constant can be valid to the displayed order along all subsequences.*

This is useful conceptually: a nonconvergent normalized ratio is not merely a nuisance in the forward asymptotics. It propagates into the inverse as a deterministic phase oscillation.

Proof [write]. The manuscript gives none. Let $\delta(x_k^{(i)}) \rightarrow \delta_i$ ($i = 1, 2$) along $x_k^{(i)} \rightarrow \infty$, with $\vartheta(q_*, \delta_1) \neq \vartheta(q_*, \delta_2)$, and take $Y = e^{G(x_k^{(i)})}$, so that $x_0 = x_k^{(i)}$. If some constant c gave $x = x_0 - (c + \tilde{H}(x_0))/G'(x_0) + o(1/G'(x_0))$ along both sequences, subtracting (19.3) would give $\log \vartheta(q(x_0), \delta(x_0)) - c = o(1)$ along both, hence $\log \vartheta(q_*, \delta_1) = c = \log \vartheta(q_*, \delta_2)$ by continuity, a contradiction. $\square$

[write] Theorem 19.1 is the first-order, real, bounded-perturbation analogue of the $M = 1$ term of the perturbed inversion theorem `p0:thm:perturbed-inversion` of the series-and-transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`, which is analytic and assumes a small perturbation parameter; neither statement contains the other.

20 Application I: high powers of ballot numbers

Part I studies

$$S_{n,k} = \sum_h B(n, h)^k,$$

where $B(n, h)$ counts nonnegative ± 1 paths of length n ending at height h . It uses

$$m = n + 1, \quad r = h + 1,$$

so r lies on a parity class with spacing $d = 2$, and the continuous peak is at $r = \sqrt{m}$. Its phase is

$$\alpha_n = \frac{\sqrt{m} - (m \bmod 2)}{2}, \quad (20.1)$$

and its theta normalization is

$$\Theta_\tau(\alpha) = \sum_{j \in \mathbb{Z}} e^{-4\tau(j-\alpha)^2}. \quad (20.2)$$

[write: this is Part I's (2.6); the manuscript writes Θ_τ^{bal} . In this section the power is Part I's k , and $r = h + 1$.]

Part I proves, uniformly for $\tau = k/m$ in compact subsets of $(0, \infty)$ (Theorem 6.2),

$$S_{n,k} = V_m^k e^{\tau/6} [\Theta_\tau(\alpha_n) + O(m^{-1/2})]. \quad (20.3)$$

This is exactly Theorem 15.1: in our convention

$$q = 8\tau, \quad \delta = \alpha_n, \quad \Omega_n = V_m e^{1/(6m)}.$$

The ballot report's local Stirling expansion verifies (H1), its global trigamma/log-concavity estimate verifies (H2), and the peak remains macroscopically far from the endpoints, giving (H3). [write: verified in Proposition 20.1; the phase α_n is used modulo 1.]

The conceptual gain is not a new proof of (20.3); that theorem is already in the repository. The gain is that its theta factor is now identified as the universal output of a reusable set of hypotheses. Moreover, the report's all-orders coefficient functions

$$\sum_{j \in \mathbb{Z}} R_s(2(j - \alpha_n)) e^{-4\tau(j - \alpha_n)^2}$$

are precisely the $\mathcal{T}_s$ of Theorem 17.2 after the change of variable $u = 2(j - \alpha_n)$. Its observed odd-order cancellation at integer and half-integer phases is Corollary 17.3. [write: the "precisely" holds only at first order with the dictionary above; see Remark 20.2.]

For the diagonal A357825, $k = n$ and hence $\tau = n/(n + 1) \rightarrow 1$. The leading normalized ratio therefore has the cluster set

$$e^{-5/6} \{\Theta_1(\delta) : \delta \in \mathcal{D}\}, \quad (20.4)$$

where $\mathcal{D}$ is the phase cluster set of (20.1). The existing ballot report analyzes this phase in detail; the transfer theorem explains why the same structure must recur in any array with the same local lattice geometry.

[write] Normalization of (20.4). The "leading normalized ratio" is Kotesovec's $a_n/\mathcal{A}_n$ of (1.3), not a_n/Ω_n^n , which by Corollary 15.2 has the cluster set $\{\Theta_1(\delta) : \delta \in \mathcal{D}\}$ without the factor. The factor comes from $\Omega_n^n/\mathcal{A}_n = (n/m)^n e^{n/(6m)} \rightarrow e^{-1} e^{1/6}$, since $V_m = K_n n/m$ by (2.4). By Proposition 4.3 the phases are dense, so $\mathcal{D} = \mathbb{R}/\mathbb{Z}$ and (20.4) is the interval $[C_-, C_+]$ of Theorem 4.2; numerically $e^{-5/6} \vartheta(8, \frac{1}{2}) = 0.319866759530995726129 \dots$ and $e^{-5/6} \vartheta(8, 0) = 0.450518194019661974351 \dots$, equal to (4.3) and (4.4).

Proposition 20.1 (The ballot array satisfies (H1)–(H3); [write]). Let $K = [8\tau_-, 8\tau_+]$ with $0 < \tau_- < \tau_+$, and let the exponent $k = \tau m$ have $\tau \in [\tau_-, \tau_+]$. In the dictionary above ($d_n = 2$, sites $r = \sqrt{m} + 2(j - \alpha_n)$, $\sigma_n^2 = m/2$), the ballot rows satisfy (H1), (H2) and (H3) on K , with $B_* = e^{2\tau_+(C+1)}$ and $c_* = 4c\tau_-$, where C, c are the constants of Lemma 3.2. Hence Theorem 15.1 recovers the leading term of (20.3), with error $o(1)$ instead of Part I's $O(m^{-1/2})$.

Proof. (H1): the display in the proof of Theorem 6.2 gives, with $u = r - \sqrt{m} = 2(j - \alpha_n)$, $k \log(b_m(\sqrt{m} + u)/V_m) = \tau(\frac{1}{6} - u^2) + O(m^{-1/2})$ on bounded windows, uniformly in τ ; subtracting $k \log e^{1/(6m)} = \tau/6$ leaves $-\tau u^2 = -\frac{\tau}{2}(j - \alpha_n)^2$ up to $O(m^{-1/2})$. (H2): Lemma 3.2 gives $n \log(b_m(r)/K_n) \leq C - cu^2$ for $0 < r \leq m$ and large n . Since $\Omega_n = K_n(n/m)e^{1/(6m)}$,

$$k \log \frac{b_m(r)}{\Omega_n} = \frac{k}{n} n \log \frac{b_m(r)}{K_n} + k \log \frac{m}{n} - \frac{\tau}{6} \leq \frac{\tau m}{n} (C - cu^2) + \tau m \log \frac{m}{m-1},$$

and $1 \leq m/n \leq 2$, $m \log \frac{m}{m-1} \leq 2$ give $k \log(b_m/\Omega_n) \leq 2\tau_+(C+1) - 4c\tau_-(j - \alpha_n)^2$ (if $C < 0$, replace C by 0). (H3): the admissible r are the integers of parity m in $(0, m]$, and the window sites $\sqrt{m} + 2(j - \alpha_n)$, $|j - \alpha_n| \leq R$, lie in that range for large m . $\square$

Remark 20.2 ("Precisely the $\mathcal{T}_s$ " holds only at first order; [write]). With the dictionary printed above ($r_n = k = n$ on the diagonal, $q_n = 8n/m$, $\Omega_n = V_m e^{1/(6m)}$, $t_n = m^{-1/2}$), the polynomials of Definition 17.1 are obtained from Lemma 3.1 with $\tau = n/m = 1 - t^2$. Written in Part I's variable $u = 2(j - \alpha_n)$, a series computation with Part I's $A_1, \dots, A_4$ gives

$$P_1 = \frac{u^3}{3} + \frac{2u}{3} = D_1, \quad P_2 = -\frac{u^4}{4} - \frac{7}{60} \neq D_2 = -\frac{u^4}{4} + u^2 + \frac{13}{60},$$

so $Q_1 = R_1$ but $Q_2 = \frac{u^6}{18} - \frac{u^4}{36} + \frac{2u^2}{9} - \frac{7}{60} \neq R_2$ (compare (2.10)). Both expansions are valid; they expand different normalized quantities. At $n = 400, 1600, 6400$ the truncation after $\mathcal{T}_2$

of a_n/Ω_n^n has relative errors $-3.4 \cdot 10^{-6}$, $-2.1 \cdot 10^{-7}$, $-1.3 \cdot 10^{-8}$, and the truncation after F_2 of $a_n/\mathcal{A}_n$ (Theorem 2.2) $7.9 \cdot 10^{-7}$, $5.0 \cdot 10^{-8}$, $3.2 \cdot 10^{-9}$, both $O(m^{-3/2})$; but $\mathcal{T}_2(q_n, \alpha_n) \approx 0.108$ while $F_2(\alpha_n) \approx 1.090$ at these n . The identification is exact at every order after re-choosing the dictionary: take $q = 8$ fixed, $t_n = m^{-1/2}$, and Ω_n with $\Omega_n^n = e^{-5/6} \mathcal{A}_n$. Then (3.7) reads $n \log(b_m(\sqrt{m} + u)/\Omega_n) = -u^2 + \sum_{s \geq 1} D_s(u)t^s$, so $P_s(v) = D_s(2v)$ in the variable $v = j - \alpha_n$, the recurrence (17.4) becomes (3.10), $Q_s(v) = R_s(2v)$, and $\mathcal{T}_s(8, \alpha) = F_s(\alpha)$ of (2.8). The display above the remark, with $e^{-4\tau(\cdots)^2}$ and $\tau \neq 1$, matches neither dictionary beyond $s = 1$.

Remark 20.3 (The inverse theorem on Part I's branches; [write]). The phase α_n jumps with the parity of m , so the integer sequence a_n has no smooth interpolation with a smooth phase, and Theorem 19.1 cannot be applied to it directly; the manuscript does not carry out the application. On each parity branch it applies to Part I's smooth model $A_{\sigma,J}$ of (8.2): take $G = g$ of (5.1), $R = 0$ and $H = \log(e^{-5/6} \sum_{j \leq J} (x+1)^{-j/2} F_j(\alpha_\sigma(x)))$, which is bounded because $\Theta \geq e^{-1}$, with $H' = O(x^{-1/2})$ because $\alpha'_\sigma(x) = 1/(4\sqrt{x+1})$; and $g'(x) = 2\lambda x + O(\log x)$, g'' bounded. Then (19.3) gives $r_{\sigma,J}(L) = x_0 - H(x_0)/g'(x_0) + o(1/x_0)$ with $g(x_0) = L$, whose theta part is $-\log \Theta(\alpha_\sigma(x_0))/g'(x_0)$, the log Θ -part of the term $-d_0(\beta)/(2\lambda s)$ of Theorem 8.2, which is sharper. A single statement for the integer thresholds across both parities is item F5 of Section 25.

21 Application II: growing powers of Mahonian coefficients

Let $I_n(j)$ be the coefficient of z^j in the z -factorial

$$[n]_z! = \prod_{m=1}^n (1 + z + \cdots + z^{m-1}),$$

and let

$$S_n(r) = \sum_j I_n(j)^r.$$

The inversion statistic has mean

$$\mu_n = \frac{n(n-1)}{4}$$

and variance

$$\sigma_n^2 = \frac{n(n-1)(2n+5)}{72}.$$

The lattice has unit spacing, while the center has phase

$$\delta_n = \mu_n - \lfloor \mu_n \rfloor \in \{0, 1/2\}.$$

A local Edgeworth expansion gives, in the central window,

$$\log \frac{I_n(j)}{I_n(\lfloor \mu_n \rfloor)} = -\frac{(j - \mu_n)^2 - \delta_n^2}{2\sigma_n^2} + \text{higher local terms.} \quad (21.1)$$

Thus the critical parameter is

$$q_n = \frac{r_n}{\sigma_n^2}.$$

When $q_n \rightarrow \rho \in (0, \infty)$, Theorem 15.1 predicts

$$\frac{S_n(r_n)}{C_n^{r_n}} \sim e^{\rho \delta_n^2/2} \sum_{x \in \mathbb{Z} + \delta_n} e^{-\rho x^2/2}, \quad C_n = I_n(\lfloor \mu_n \rfloor), \quad (21.2)$$

which is exactly the theta normalizer used in the repository’s Mahonian growing-power report. [write: the manuscript’s $M_n(j)$, q and V_n are renamed $I_n(j)$, z and σ_n^2 (Table 2); the Mahonian report writes $M_n(k)$ and V_n . The right side of (21.2) is that report’s $\Theta_\delta^{\text{mgp}}(\rho)$; its phase convention $\{0, \frac{1}{2}\}$ differs from Definition 14.1’s $[-\frac{1}{2}, \frac{1}{2})$ only by an integer, which changes nothing.]

The public OEIS entry A380275 records the fourth-power sum and now cites Wang’s proof of the fixed-real-power asymptotic. Wang’s preprint explicitly keeps r fixed and states no uniformity in r ; the critical scaling $r \asymp \sigma_n^2 \asymp n^3$ is therefore a genuinely different asymptotic regime. Canfield, Janson and Zeilberger provide classical central and local-normal background for Mahonian statistics [16]. The transfer theorem isolates the additional fact needed for growing powers: after the local quadratic law is multiplied by r_n , the unit lattice no longer disappears.

21.1 A two-instance universality statement

The ballot and Mahonian rows have very different combinatorial origins and different peak scales:

	Ballot endpoints	Mahonian inversions
continuous center	$\sqrt{n+1}$ (shifted variable)	$n(n-1)/4$
lattice span	2	1
curvature scale	order $\sqrt{n}$ in endpoint coordinate	$\sigma_n \asymp n^{3/2}$
critical power	$k \asymp n$	$r \asymp \sigma_n^2 \asymp n^3$
phase	drifting square-root/parity phase	0 or $1/2$
leading critical object	translated theta sum	translated theta sum

After normalization, both reduce to the same datum (q, δ) in Theorem 15.1. This is the precise sense in which Q6 has a positive answer.

[write] The manuscript’s “predicts” is accurate: it does not verify (H1)–(H3) for the Mahonian rows, and its abstract and contribution (vi) say more. The leading statement is proved below, with (H2) weakened to (H2e). Its ingredients are the Mahonian report’s own (the cancellation-respecting profile estimate `mgp:eq:profileleading` and log-concavity of the row), so the proposition is a reorganization of that report’s proof of its critical window, not an independent second proof.

Proposition 21.1 (The Mahonian critical window is an instance; [write]). Take $\xi_n = 0$, $d_n = 1$, $x_n = \mu_n$ (so $\delta_n \equiv \mu_n \pmod{1}$, $\delta_n \in \{0, -\frac{1}{2}\}$), σ_n^2 as above and $\Omega_n = C_n e^{\delta_n^2/(2\sigma_n^2)}$. For every compact $K \subset (0, \infty)$ and exponents with $q_n = r_n/\sigma_n^2 \in K$, the Mahonian rows satisfy (H1), (H2e) with $B_* = e^{1+3q_-}$, $c_* = q_-$, and (H3). Consequently

$$\frac{S_n(r_n)}{C_n^{r_n}} = e^{q_n \delta_n^2/2} \mathfrak{g}(q_n, \delta_n) + o(1) = \Theta_\delta^{\text{mgp}}(q_n) + o(1),$$

uniformly for $q_n \in K$: the critical-window case of the Mahonian report’s Theorem `mgp:thm:main`.

Proof. The sites are $\mu_n + v$ with $v = j - \delta_n \in \mathbb{Z} + \delta$. In the Mahonian report, $I_n(\mu_n + v) = n! f_n(v)$ with f_n even, and $C_n = n! f_n(\delta)$ (for $\delta = \frac{1}{2}$ by the symmetry $I_n(k) = I_n(N - k)$, $N = 2\mu_n$). Its estimate `mgp:eq:profileleading`, $\log(f_n(v)/f_n(\delta)) = -\frac{v^2 - \delta^2}{2\sigma_n^2} \{1 + \mathcal{O}_B(n^{-1})\}$ for $|v| \leq B\sigma_n$ and $|v| \geq \delta$, gives on $|v| \leq R$

$$r_n \log \frac{I_n(\mu_n + v)}{\Omega_n} = -\frac{q_n}{2} v^2 + \frac{q_n}{2} (v^2 - \delta^2) \mathcal{O}_R(n^{-1}),$$

whose error is at most $q_+ R^2 \mathcal{O}_R(n^{-1}) \rightarrow 0$: this is (H1). The window sites lie in the support $[0, N]$, $N = n(n-1)/2$, for large n , which is (H3); every $I_n(k)$ with $0 \leq k \leq N$ is positive. The Mahonian report proves that the row is log-concave (its Section 3), so Proposition 15.5 gives (H2e) and the transfer. Finally $\Omega_n^{r_n} = C_n^{r_n} e^{q_n \delta_n^2/2}$, and (12.1) converts $e^{q \delta^2/2} \vartheta(q, \delta)$ to $\Theta_\delta^{\text{mgp}}(q)$. $\square$

A numerical check on exact rows ($n = 40, 41, 80, 82$; $q_n = 0.5, 1, 3$) found the rows log-concave and the bound (15.5) with these constants satisfied with room (the largest ratio of a powered summand to the bound was 0.106), and window errors $\eta_n(3)$ from 0.36 at $n = 40$, $q_n = 3$ down to 0.020 at $n = 82$, $q_n = 0.5$, decreasing like $1/n$. Whether the Mahonian rows satisfy the Gaussian (H2) itself is item F1 of Section 25.

22 A practical verification checklist for new OEIS arrays

For future applications, the theorem suggests a short proof architecture.

- Step 1. Find a continuous carrier.** Extend the row locally to a smooth function and locate its peak x_n . Record the lattice spacing and phase before replacing x_n by a rounded mode.
- Step 2. Measure curvature.** Compute $\sigma_n^2 = -1/(\partial_x^2 \log C_n)(x_n)$ to leading order. The critical power is $r_n \asymp \sigma_n^2/d_n^2$.
- Step 3. Expand the powered logarithm, not merely the row.** The right error criterion is r_n times the local logarithmic error. An error negligible for fixed powers may be leading at critical powers.
- Step 4. Prove a global summable majorant.** Log-concavity, an exact adjacent ratio, a characteristic-function estimate, or a large-deviation bound can supply (H2). A local Edgeworth expansion alone is not enough. *[write: log-concavity supplies (H2e), which suffices for the leading theorem (Proposition 15.5), not (H2) itself.]*
- Step 5. Keep the phase.** Express sites as $x_n + d_n(j - \delta_n)$. Do not round the peak until after the theta sum is visible.
- Step 6. For all orders, expand the logarithm first.** Insert the resulting P_ℓ into recurrence (17.4); this avoids repeated hand expansion of exponentials.
- Step 7. For inversion, transport the theta factor explicitly.** Apply Theorem 19.1 rather than replacing an oscillatory amplitude by a fitted constant.

23 Further deductions from the abstract theorem

23.1 Phase averaging

Suppose the phases δ_n are equidistributed modulo 1 and $q_n \rightarrow q > 0$. Then for every continuous test function φ ,

$$\frac{1}{N} \sum_{n \leq N} \varphi\left(\frac{S_n(r_n)}{\Omega_n^{r_n}}\right) \longrightarrow \int_0^1 \varphi(\vartheta(q, \delta)) d\delta, \quad (23.1)$$

provided the equidistribution is known independently. In particular,

$$\int_0^1 \vartheta(q, \delta) d\delta = \sqrt{\frac{2\pi}{q}}. \quad (23.2)$$

Thus the continuous Gaussian amplitude reappears as the *phase average* of the critical lattice law. This identity cleanly relates the subcritical Riemann-sum picture to the critical theta picture.

Proof. The first claim is Weyl equidistribution plus Theorem 15.1 and continuity of $\varphi \circ \vartheta$. For (23.2), integrate termwise and use the translates of $[0, 1]$ to tile $\mathbb{R}$:

$$\int_0^1 \sum_j e^{-q(j-\delta)^2/2} d\delta = \int_{\mathbb{R}} e^{-qx^2/2} dx.$$

□

[write] At $q = 8$ this is Part I's $\int_0^1 \Theta(\alpha) d\alpha = \sqrt{\pi}/2$ in the proof of Corollary 4.4, where the equidistribution is Proposition 4.3.

23.2 A quantitative phase sensitivity bound

Differentiating the theta sum gives

$$\partial_\delta \vartheta(q, \delta) = q \sum_j (j - \delta) e^{-q(j-\delta)^2/2}.$$

For q in a compact interval $K \subset (0, \infty)$, the derivative is uniformly bounded. Therefore a phase approximation $\widetilde{\delta}_n$ with $|\widetilde{\delta}_n - \delta_n| = o(1)$ changes the leading term only by $o(1)$. In contrast, replacing δ_n by a fixed value with an $O(1)$ phase error can change the leading constant by an $O(1)$ amount.

23.3 Mode multiplicity as a large- q invariant

Proposition 16.1 shows that in the supercritical theta model the number of nearest lattice points becomes the leading discrete multiplicity. A generic phase has one nearest site; a half-lattice phase has two. This explains the ubiquitous logistic/two-mode transition near exact ties in concrete high-power sums. [write: the last sentence is a heuristic, item F4 of Section 25. Part I's logistic transition (Corollary 6.4) is proved from the exact entry ratio, outside the critical window, not from Proposition 16.1.]

24 Limits of the theorem

The result resolves the repository question in the sense of giving broad, checkable sufficient hypotheses. Several boundaries should remain explicit.

1. The theorem is not a characterization. Arrays with weaker tail control or nonquadratic peaks may still have theta limits.
2. Degenerate peaks are outside scope. If the first nonzero local term is quartic or higher, the critical lattice sum is no longer Gaussian-theta; a different universal kernel appears.

3. Boundary peaks require a half-lattice or cone treatment. The symmetric whole-lattice extension used here is then invalid.
4. Multiple separated peaks produce a sum of theta sectors with relative exponential weights and possible Stokes-like switching as parameters vary.
5. The all-orders theorem is Poincaré, with fixed truncation order. No convergence, optimal truncation, Borel summability, or resurgence is asserted.
6. The theorem says nothing arithmetic about the exact integers. In particular it does not address the prime-power congruence conjecture recorded on OEIS A357825.
7. No statement in this article is formalized in Lean or another proof assistant, and no publication-priority claim is made.

[write: the assertions in items 2–4 about what happens outside the scope are not proved; item F4 of Section 25. Item 6 is Part I's question Q9.]

25 Further questions and research

The transfer theorem suggests a concrete continuation program.

- R1. Necessary and sufficient conditions.** Replace the sufficient local-plus-tail hypotheses by a characterization in terms of convergence of the powered row as a point process on the peak lattice.
- R2. Degenerate saddles.** Develop the analogue for quartic and higher even-order maxima. The expected critical objects are lattice sums of $\exp(-c|j - \delta|^{2m})$, with new phase diagrams.
- R3. Boundary and cone theta laws.** Treat peaks approaching a hard boundary at the same scale as the powered width; in several dimensions this should produce theta sums over cones or affine semigroups.
- R4. Multiple peaks and switching.** Give a uniform theorem for finitely many competing peaks and locate the parameter surfaces on which dominant theta sectors exchange order.
- R5. Growing-order asymptotics.** Bound the growth of the Q_m from (17.4) and identify conditions under which optimal truncation is possible.
- R6. Exponential completion.** After optimal algebraic truncation, identify secondary lattice peaks or complex saddles that generate exponentially small sectors.
- R7. Certified finite thresholds.** Make the Gaussian majorant and local remainder constants explicit enough to certify the first n at which a chosen asymptotic bracket holds.
- R8. Multidimensional OEIS cases.** Search arrays indexed by two or more endpoint statistics for critical growing powers whose limiting multiplier is a nontrivial quadratic-form theta function.
- R9. Coxeter-group Mahonian arrays.** Combine local limit theorems for inversion statistics in other finite Coxeter groups with Theorem 18.1; determine which root lattices and phase classes occur.
- R10. Phase equidistribution.** For square-root, algebraic, or logarithmic peak locations arising in concrete arrays, prove discrepancy estimates strong enough to obtain effective versions of (23.1).

- R11. Inverse staircase certification.** Combine Theorem 19.1 with monotone interpolation bounds to turn oscillatory real inverses into exact ceiling/floor statements for integer thresholds.
- R12. Formalization.** Formalize the finite-window dominated-convergence core first, then a reusable Gaussian lattice-tail lemma, and finally the one-dimensional transfer theorem. This would create a small analytic library usable by several OEIS developments.

[write] Relation to Part I's questions. R5 and R6 are the general form of Q4, R7 of Q1, R10 of Q8 and R12 of Q10 (Section 10). For R11, the repository's only related formal statement is the generic staircase lemma `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean` (a ceiling is determined when an uncertainty interval does not straddle an integer), which concerns an arbitrary real number, not these sums.

Claims of the manuscript that are not proved (standing rule). *[write, 5 October 2026.]* Each is stated as an open question, with its source in the manuscript, its argument sketch and what is missing.

- F1. Gaussian domination for the Mahonian rows.** Source: the abstract and contribution (vi) ("the ballot and Mahonian theorems fit the hypotheses"), Section 21. The write proves (H1), (H2e), (H3) (Proposition 21.1), which suffices for the leading theorem. Open: does (H2) hold, uniformly for $q_n \in K$? Sketch: on $|v| \leq B\sigma_n$ the profile estimate gives a Gaussian bound; beyond, one needs a large-deviation bound $\log(I_n(\mu_n+v)/C_n) \leq -c v^2/\sigma_n^2$ uniformly up to the edge of the support, i.e. a quadratic lower bound on the rate function of the inversion count that is uniform in n . Missing: that uniform bound, and for Theorem 17.2 its polynomially weighted form.
- F2. Outside the critical window.** Source: Remark 16.2. Open: which hypotheses give the $q_n \rightarrow 0$ and $q_n \rightarrow \infty$ limits (16.1), (16.2) for the power sums themselves? The sketch (local expansion on $\asymp q_n^{-1/2}$ sites; errors $o(1/q_n)$ at the nearest sites) is a heuristic; no theorem is stated. The Mahonian report proves both adjacent regimes for its array (Theorem `mgp:thm:main`, cases 1 and 3), Part I the supercritical one for the ballot array (Theorem 6.3).
- F3. Constrained and boundary variants.** Source: Remark 18.2, Section 24 item 3. Open: the claimed reduction of sublattice and conservation-law peaks, and the claim that boundary peaks lead to half-theta or cone-theta functions; overlaps R3. Missing: statement and proof.
- F4. Heuristics about other regimes.** Source: Section 24 items 2–4 ("a different universal kernel appears" at degenerate peaks; "Stokes-like switching" between separated peaks) and Section 23.3 ("this explains the ubiquitous logistic/two-mode transition"). Open: overlaps R2 and R4; for the last, a theorem deriving a logistic law near ties from a supercritical lattice-peak hypothesis, of which Part I's Corollary 6.4 would be an instance. Missing: any such statement.
- F5. The inverse theorem for A357825 across parities.** Source: Section 19 with Section 20 (a nonconvergent ratio "propagates into the inverse"). Remark 20.3 applies Theorem 19.1 on each parity branch. Open: a first-order inverse statement for interpolations whose phase is only piecewise smooth (switching at a sequence of points), which would cover the integer thresholds $N(y)$ of Part I's Section 8 without choosing a branch; overlaps R11.
- F6. The case $M' = 0$ of Definition 17.1.** Source: the sentence "Enlarging M' if necessary is harmless". The write's completion of Theorem 17.2 uses $M' > 0$. Open: does the theorem

hold with $M' = 0$? It does if additionally $c_* < q_-/2$; missing is a proof that the hypotheses force this, or a counterexample.

F7. The manuscript’s (H2) for the ballot array at all orders. Source: Section 20 (“all-orders coefficient functions are precisely the $\mathcal{T}_s$ ”). With the re-chosen dictionary of Remark 20.2 the coefficients agree; Part I proves the expansion directly. Open: a check that Definition 17.1, including (17.2) and (17.3), holds for the ballot rows, so that Theorem 17.2 re-derives Theorem 2.2 rather than merely matching it. Sketch: Part I’s window $|u| \leq m^{1/12}$ and Lemma 3.2 with polynomial weights. Missing: the bookkeeping.

26 Conclusion

The common structure behind the ballot and Mahonian OEIS examples is now explicit. A positive row has a moving smooth peak; taking a growing power narrows it; when the narrowed width reaches the lattice spacing, the lattice phase survives in the leading term. Under a local quadratic law and a global Gaussian majorant, the outcome is universally a translated theta function. Higher local terms pass through a universal Bell-polynomial mechanism and become theta derivatives, while several endpoint coordinates produce quadratic-form theta series.

This resolves the specific open request for a reusable lattice-peak theorem in the ProveIt OEIS collection. It also clarifies the neighboring regimes and inversion: the ordinary Gaussian constant is the phase average of the critical theta law, exact or near ties control the supercritical limit, and the same phase-dependent multiplier shifts inverse thresholds by an explicit Newton term. The theorem therefore supplies a common analytic interface between local limit theory, discrete Laplace asymptotics, OEIS high-power sums, and transseries-style inversion.

[write] “Resolves” is meant in the sufficient-hypothesis sense of Section 24; the “neighboring regimes” are clarified at the level of the theta function (Proposition 16.1), not proved for power sums (item F2).

A Additional explicit coefficients

For direct implementation, the next two multiplier polynomials are

$$R_3(u) = \frac{u^9}{162} - \frac{5u^7}{108} + \frac{119u^5}{270} + \frac{197u^3}{1620} - \frac{7u}{18}, \quad (\text{A.1})$$

$$R_4(u) = \frac{u^{12}}{1944} - \frac{19u^{10}}{1944} + \frac{1429u^8}{12960} - \frac{3017u^6}{9720} + \frac{5227u^4}{19440} + \frac{311u^2}{540} + \frac{8263}{50400}. \quad (\text{A.2})$$

The next two Stirling coefficient polynomials are

$$A_4(x) = -\frac{x^{10}}{90} + \frac{x^8}{8} - \frac{x^6}{3} + \frac{4x^2}{15}, \quad (\text{A.3})$$

$$A_5(x) = -\frac{x^{12}}{132} + \frac{x^{10}}{10} - \frac{x^8}{3} + \frac{2x^4}{3} - \frac{1}{20}. \quad (\text{A.4})$$

The archive supplies D_j and R_j through $j = 6$ in `data/coefficients.json`. The source function `symbolic(order)` can generate higher specified orders; its numerical evaluator is

intentionally configured for the supplied orders 0 through 6 and rejects unsupported numerical truncations.

The following identities can be used instead of differentiating theta series numerically:

$$M_1 = \frac{\Theta'}{4}, \quad M_2 = \frac{\Theta''}{16} + \frac{\Theta}{2}, \quad M_3 = \frac{\Theta'''}{64} + \frac{3\Theta'}{8}.$$

For τ near 1, direct Gaussian sums are especially efficient. For small τ , the Fourier representation (4.1) is preferable because its nonzero modes are exponentially suppressed. Whichever representation is used, rigorously certified computation requires an explicit tail bound and rounding control; the numerical script uses high precision but does not claim interval certification.

B A compact dependency and trust map

The logical dependency chain is:

$$\text{ballot identity} \longrightarrow \begin{cases} \text{Stirling expansion + concavity} \longrightarrow \text{theta expansion,} \\ \text{exact entry ratio} \longrightarrow \text{two-site theorem,} \\ \text{rising factorial} \longrightarrow \text{multiset sectors.} \end{cases}$$

The theta expansion plus elementary square-root phase counting proves the limiting interval, distribution, and quotient statements. Smooth finite model inversion proves the inverse results. The multiset-sector proof also uses the uniform critical estimate. No finite numerical test is an assumption in these implications.

The exact computations test definitions and coefficient algebra; the numerical tables test consequences. Neither is a premise of an analytic theorem. The standard gamma and theta inputs are cited explicitly. No conjecture from ProveIt or OEIS is imported as a lemma.

C Elementary analytic lemmas (Part II)

[write] This appendix and the next are Appendices A and B of the manuscript of Part II.

Lemma C.1 (Uniform translated Gaussian tails). *For every $c > 0$,*

$$\sup_{\delta \in [-1/2, 1/2]} \sum_{|j-\delta| > R} e^{-c(j-\delta)^2} \longrightarrow 0 \quad (R \rightarrow \infty).$$

Proof. For $R \geq 1$ and $|j - \delta| > R$, one has $|j| > R - 1/2$. Hence the left side is bounded by $2 \sum_{m > R-1/2} \exp[-c(m - 1/2)^2]$, a convergent tail independent of δ . $\square$

Lemma C.2 (Continuity of theta in scale and phase). *The series (14.3) and all mixed derivatives in (q, δ) converge locally uniformly on $(0, \infty) \times \mathbb{R}$. In particular ϑ is smooth there and periodic in δ .*

Proof. On a compact $q \in [q_-, q_+]$, any derivative is a polynomial in $j - \delta$ times $\exp[-q(j - \delta)^2/2]$. Uniform convergence follows from Lemma C.1 with a polynomially weighted variant, which is proved identically. $\square$

D Source and novelty audit (Part II)

The repository facts used to motivate the problem were checked against the ProveIt default branch on 5 October 2026. The A357825 report explicitly lists Q6, “A reusable lattice-peak theorem for other path models,” and the A380274 report explicitly says that its Mahonian theorem is another instance but does not prove the general result. Public checks of OEIS A357825 and A380275 confirm the sequence definitions and the currently displayed asymptotic comments. Wang’s 2026 Mahonian preprint states a fixed-real-power expansion and explicitly disclaims uniformity in the power. Canfield–Janson–Zeilberger supply classical asymptotic-normal and local-limit background. Helton–Hughes–Schlosser give a recent general discrete Laplace framework in a fine-lattice regime.

No exhaustive worldwide novelty search is claimed. The mathematically precise claim of this article is narrower: the theorems above are proved here and answer the *repository’s* Q6 by giving explicit sufficient hypotheses that simultaneously cover its ballot and Mahonian examples. The standard ingredients—dominated convergence, Gaussian tails, Poisson summation, local expansions, and Newton inversion—are not claimed as new methods.

[Write note, batch 98D, 5 October 2026.] The manuscript’s bibliography gives two OEIS titles in paraphrase (“power-sum array of semi-Dyck path endpoint counts” for A357824, “fourth powers of Mahonian coefficients” for A380275). The entries’ names, fetched by the write on 5 October 2026, are “Total number $A(n,k)$ of k -tuples of semi-Dyck paths from $(0,0)$ to $(n,n-2*j)$ for $j=0..\text{floor}(n/2)$; square array $A(n,k)$, $n \geq 0$, $k \geq 0$, read by antidiagonals” (A357824; Part I’s [3] paraphrases it too) and “Sum of the fourth powers of the coefficients of q in the q -factorials” (A380275, [14]). OEIS text is licensed CC BY-SA 4.0; no OEIS data is shipped with Part II. The write did not re-fetch Wang’s preprint or the Helton–Hughes–Schlosser paper; the latter is Part I’s [8]. The claim that the hypotheses “simultaneously cover” the Mahonian example holds, after the write, in the form of Proposition 21.1. The manuscript’s own bibliography entry for this report is omitted: it is this report.

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